

Chapter 1

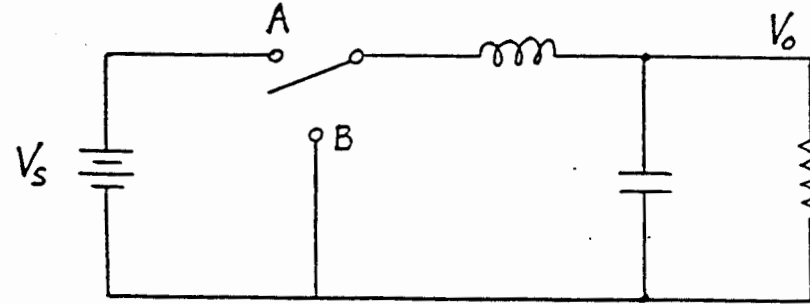
Average Model of PWM Converters

Chapter 1.1

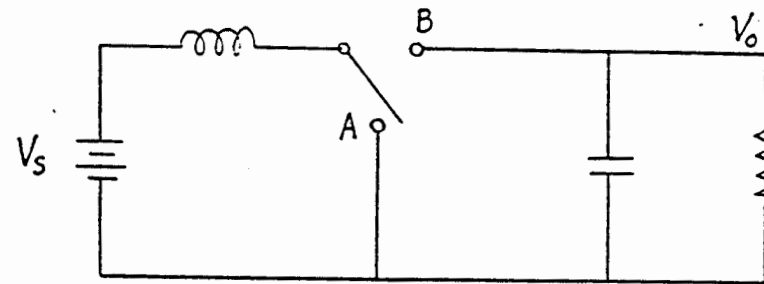
DC Analysis of PWM Converters

DC Analysis of PWM Converters

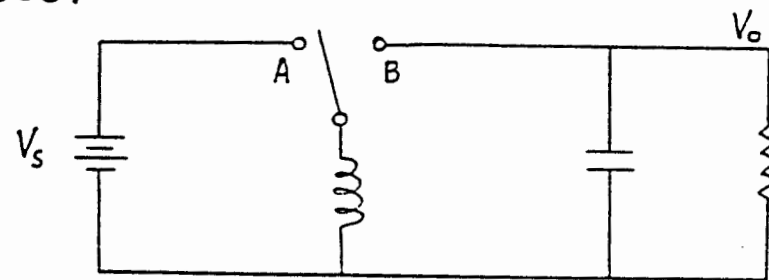
BUCK



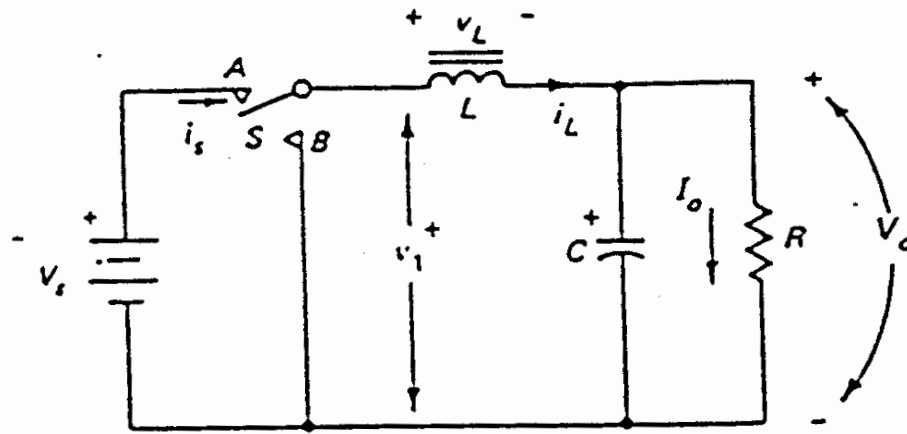
BOOST



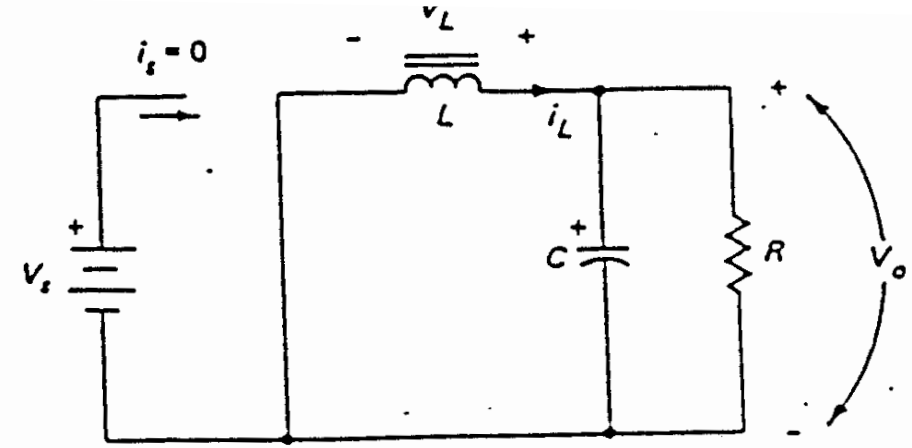
BUCK-BOOST



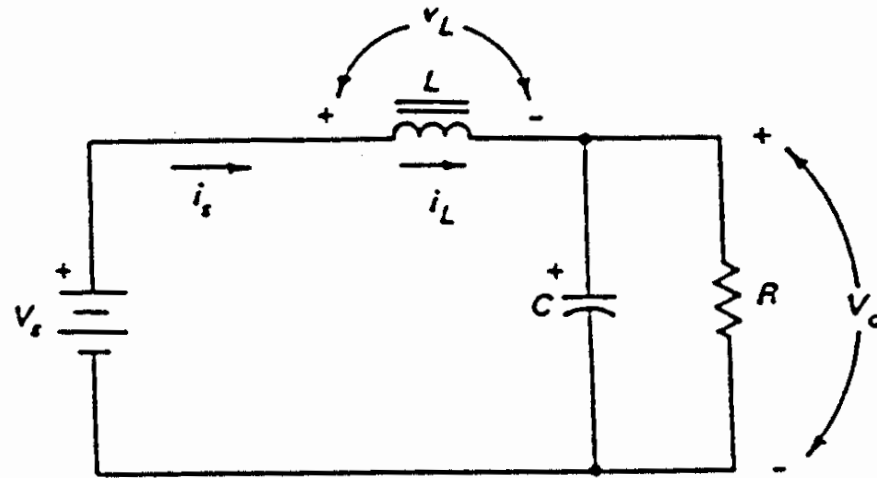
Buck Converter DC Analysis



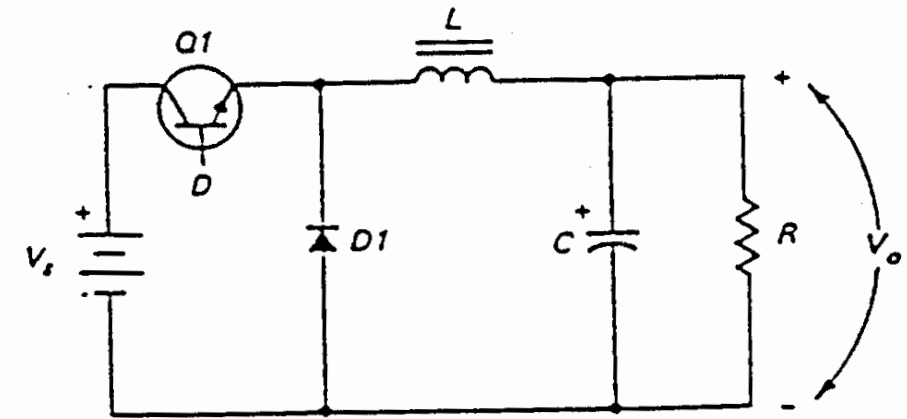
(A)



(C)

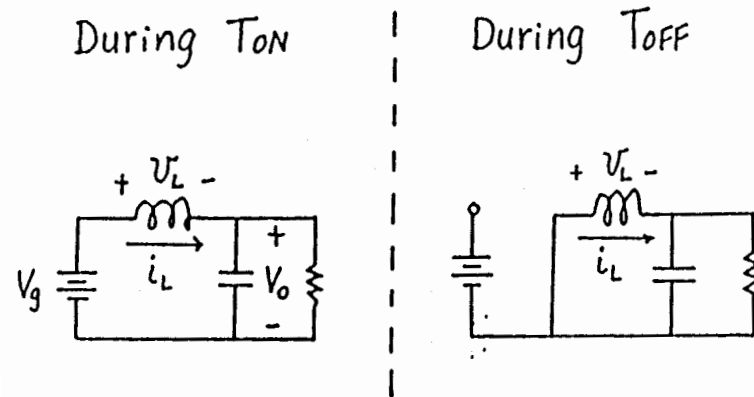


(B)



(D)

Steady State Analysis



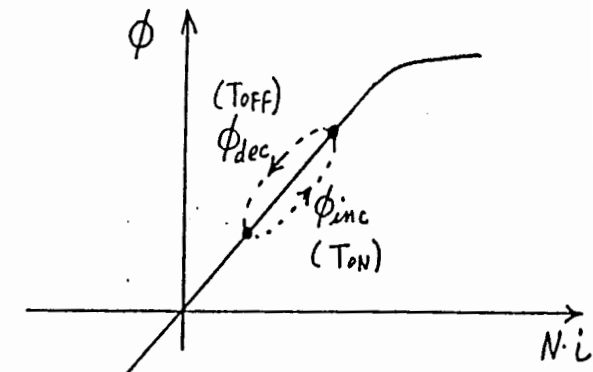
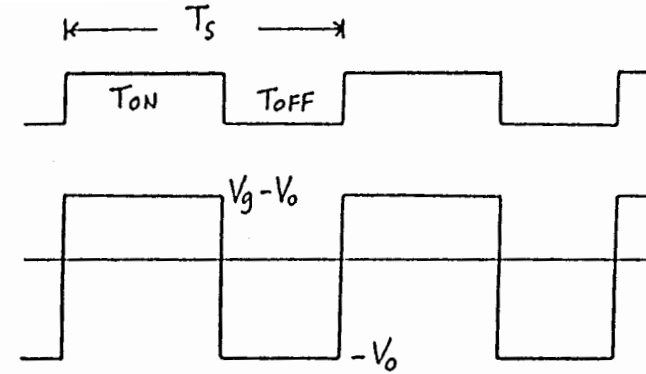
$$v_L = N \frac{d\phi}{dt}$$

$$\begin{aligned} \Delta\phi_{inc} &= \left| \frac{v_L}{N} \cdot \Delta t \right| \\ &= \frac{V_g - V_o}{N} \cdot T_{ON} \end{aligned}$$

$$\begin{aligned} \Delta\phi_{dec} &= \left| \frac{v_L}{N} \cdot \Delta t \right| \\ &= \frac{V_o}{N} \cdot T_{OFF} \end{aligned}$$

- Flux (volt-sec) Balance

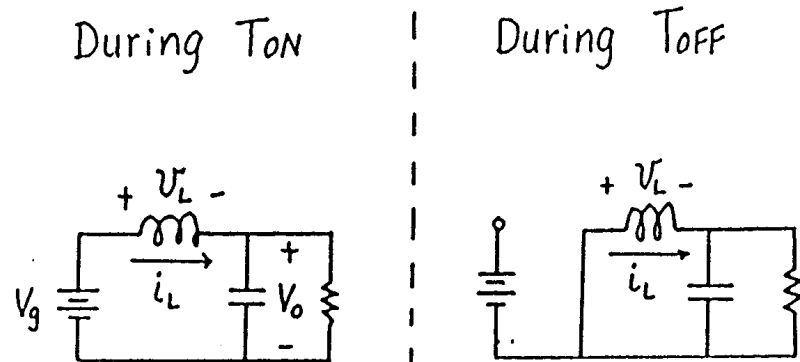
$$\Delta\phi_{inc} = \Delta\phi_{dec}$$



$$\Rightarrow (V_g - V_o) \cdot T_{ON} = V_o \cdot T_{OFF}$$

$$\therefore V_o / V_g = T_{ON} / T_s \triangleq D$$

Steady State Analysis



$$v_L = L \frac{di_L}{dt}$$

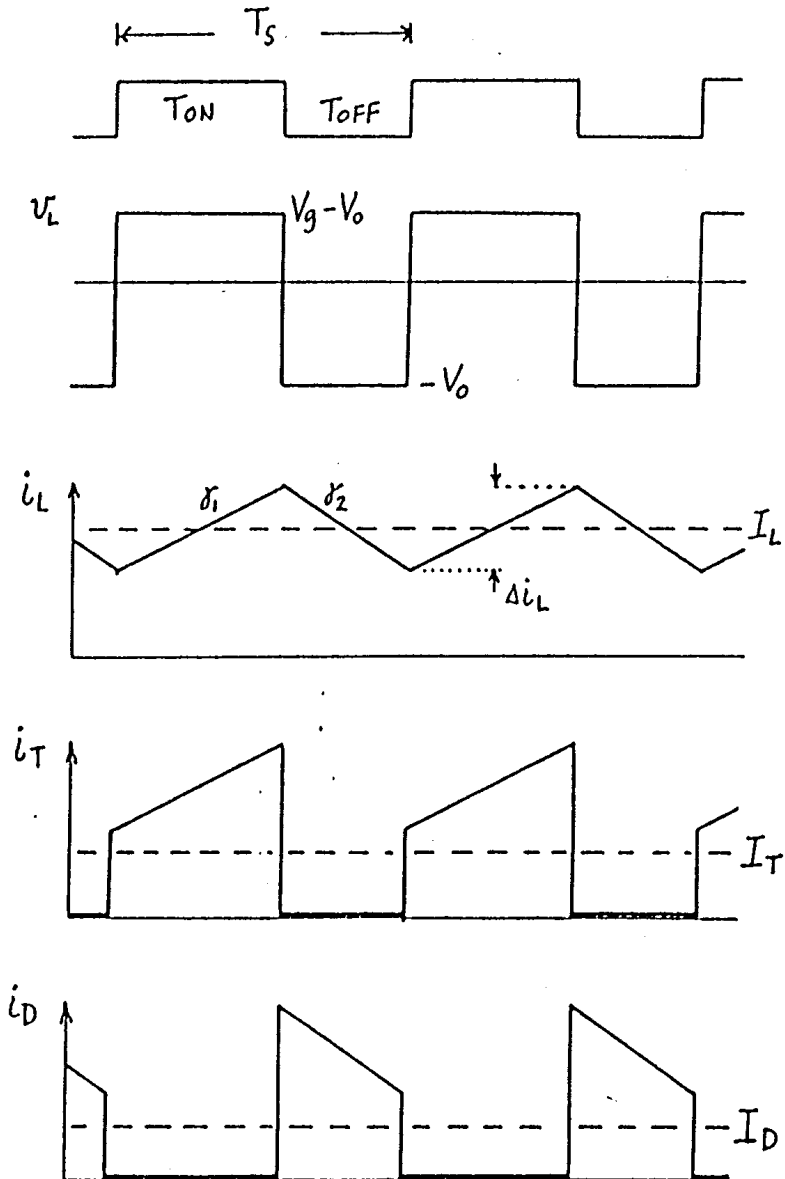
$$\Delta i_L = \frac{V_g - V_o}{L} \cdot T_{ON}$$

$$\Delta i_L = \frac{V_o}{L} \cdot T_{OFF}$$

$$\bullet I_L = I_o = \frac{V_o}{R}, \quad I_{peak} = I_L + \frac{1}{2} \Delta i_L$$

$$\bullet I_s = D \cdot I_o = I_T$$

$$\bullet I_D = I_L - I_T = D' I_o, \quad D' = 1 - D = \frac{T_{OFF}}{T_s}$$



CCM/DCM Boundary Condition

At the boundary, $\frac{1}{2} \Delta i_L = I_L$

$$I_L > \frac{1}{2} \Delta i_L \quad ; \text{ CCM}$$

$$I_L < \frac{1}{2} \Delta i_L \quad ; \text{ DCM}$$

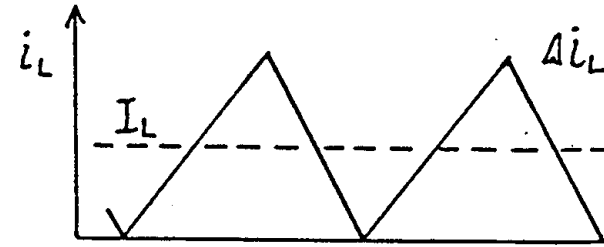
$$\Delta i_L = \frac{V_o}{L} D' T_s$$

$$I_L = \frac{V_o}{R}$$

The boundary Condition

$$\frac{1}{2} \cdot \frac{V_o}{L} D' T_s = \frac{V_o}{R}$$

or $\boxed{\frac{L}{R T_s} \Big|_{\text{crit}} \triangleq \tau_{L \text{crit}} = \frac{D'}{2}}$



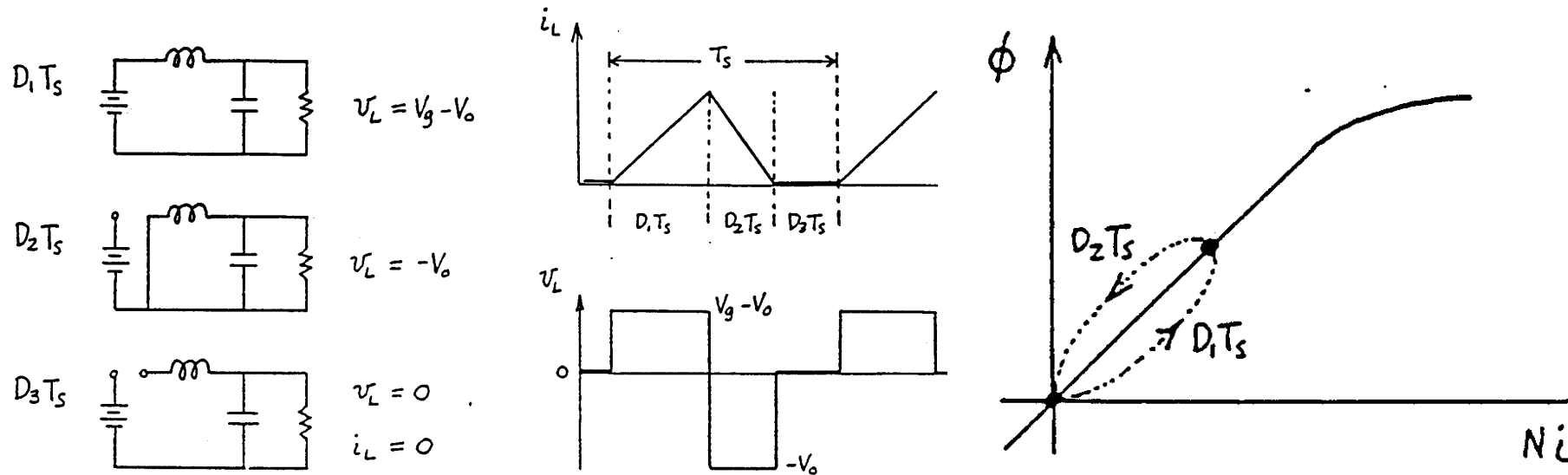
CCM if :

$$R < R_{\text{crit}} = \frac{2L}{D' T_s}$$

$$L > L_{\text{crit}} = \frac{D' R T_s}{2}$$

$$T_s < T_{s \text{crit}} = \frac{2L}{D' R}$$

Discontinuous Conduction Mode (DCM)



• Flux Balance

$$(V_g - V_o) \cdot D_1 T_s = V_o D_2 T_s$$

$$\Rightarrow \frac{V_o}{V_g} \triangleq M = \frac{D_1}{D_1 + D_2} \quad \text{--- ①}$$

* $\begin{cases} D_1 ; \text{Controllable (known)} \\ D_2 ; \text{Circuit determined} \end{cases}$
Need to find $D_2 = f(D_1)$ or $M = f(D_1)$

Discontinuous Conduction Mode (DCM)

$$I_L = I_o = \frac{V_o}{R} \quad \text{--- ②}$$

Also,

$$\begin{aligned} I_L &= \frac{1}{2} (\Delta i_L) (D_1 + D_2) \\ &= \frac{1}{2} \left(\frac{V_g - V_o}{L} \cdot D_1 T_s \right) (D_1 + D_2) \quad \text{--- ③} \end{aligned}$$

Set ② = ③ and divide both side by V_g ,
and substitute $\tau_L \triangleq \frac{L}{RT_s}$

$$\frac{1}{2} \frac{(1-M)}{\tau_L} D_1 \cdot (D_1 + D_2) = M$$

$$\Rightarrow \frac{D_1 (D_1 + D_2)}{2 \tau_L} = \frac{M}{1-M} = \frac{D_1}{D_2} \quad \text{--- ④}$$

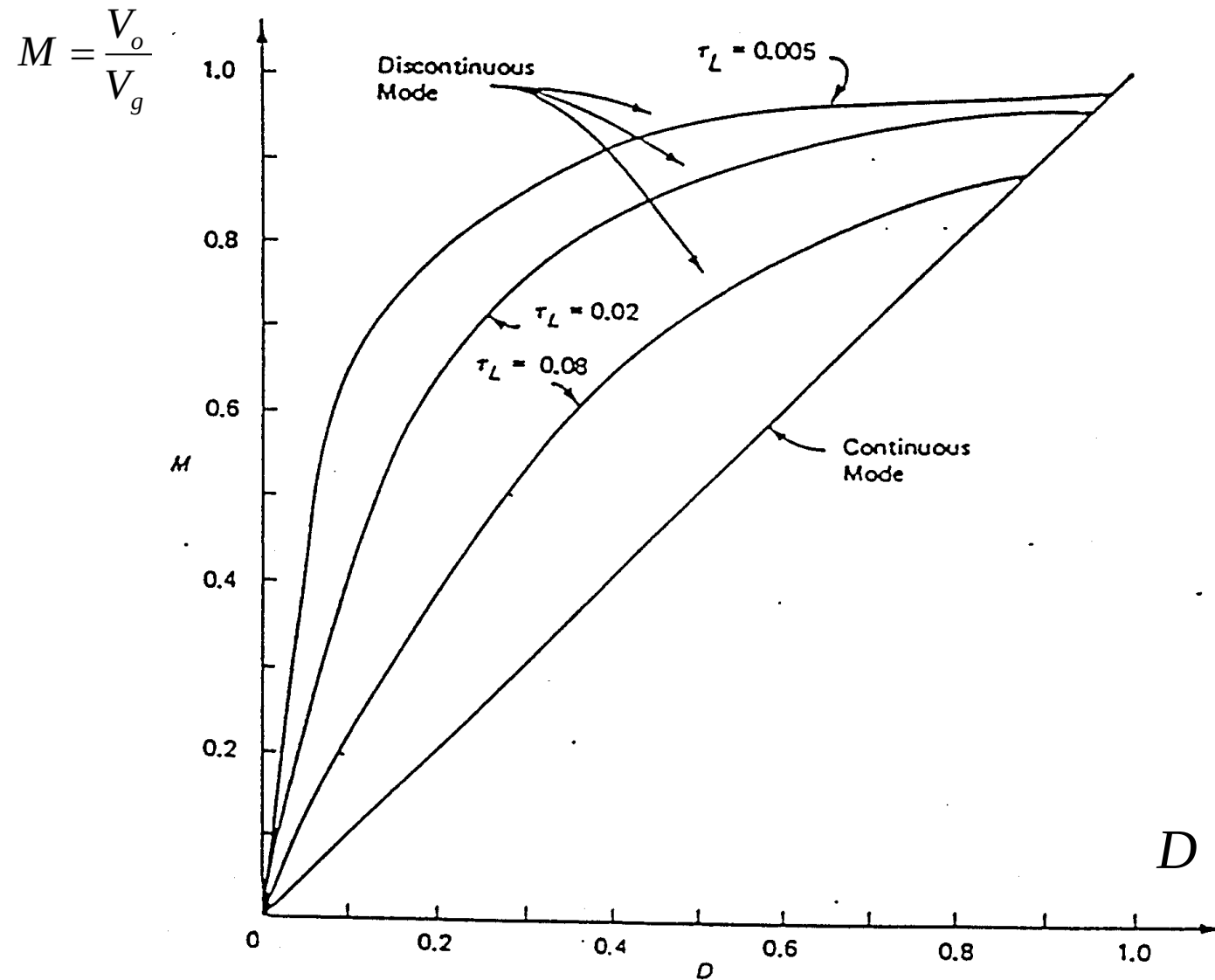
$$\Rightarrow D_2^2 + D_1 D_2 - 2 \tau_L = 0$$

$$\therefore D_2 = \frac{D_1}{2} \left(-1 + \sqrt{1 + \frac{8 \tau_L}{D_1^2}} \right) \quad \text{--- ⑤}$$

Substitute ⑤ to ①

$$M = \frac{2}{1 + \sqrt{1 + \frac{8 \tau_L}{D_1^2}}}$$

DC Gain Characteristic of BUCK Converter



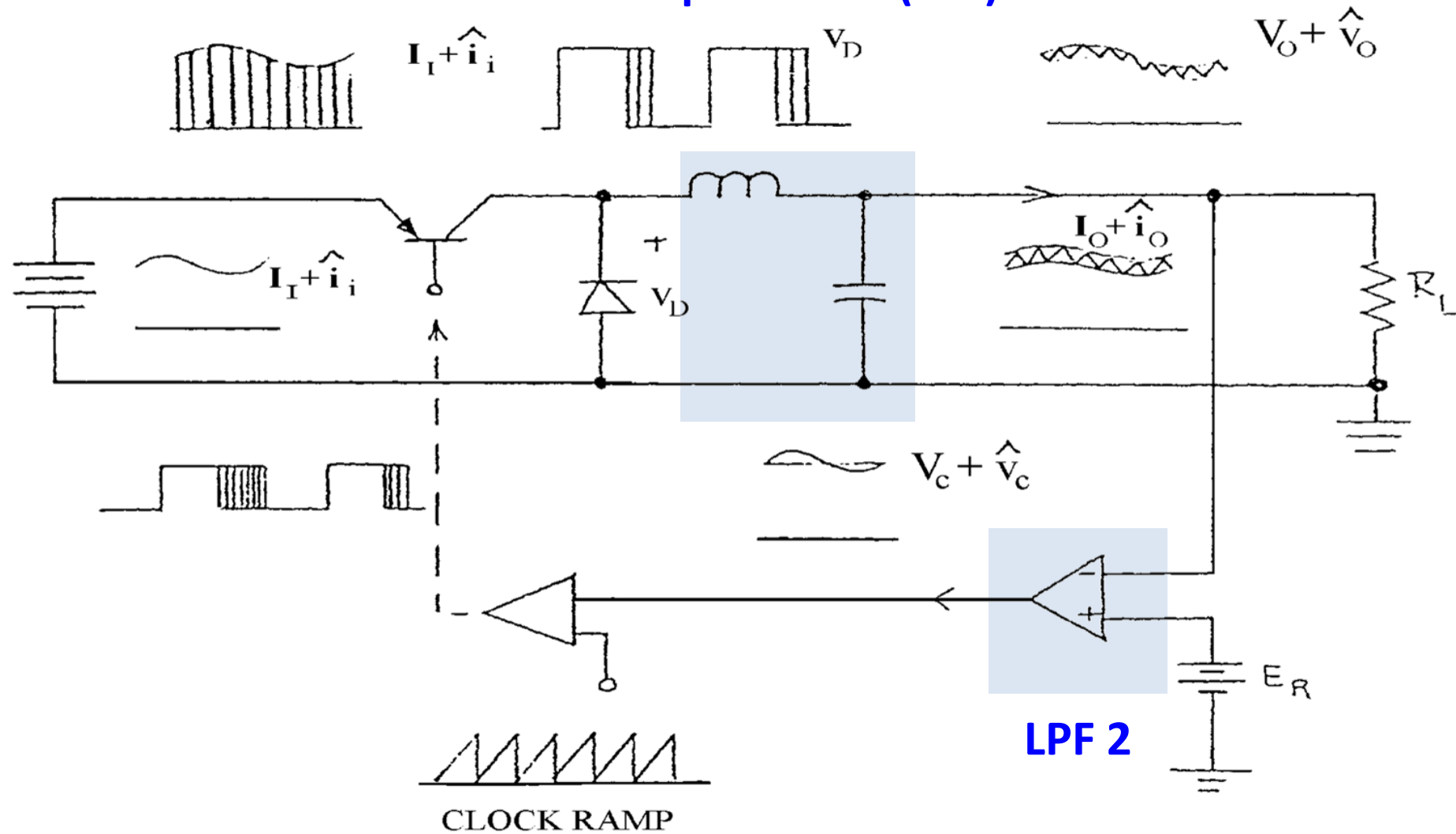
Chapter 1.2

Basic Concept of Average Model

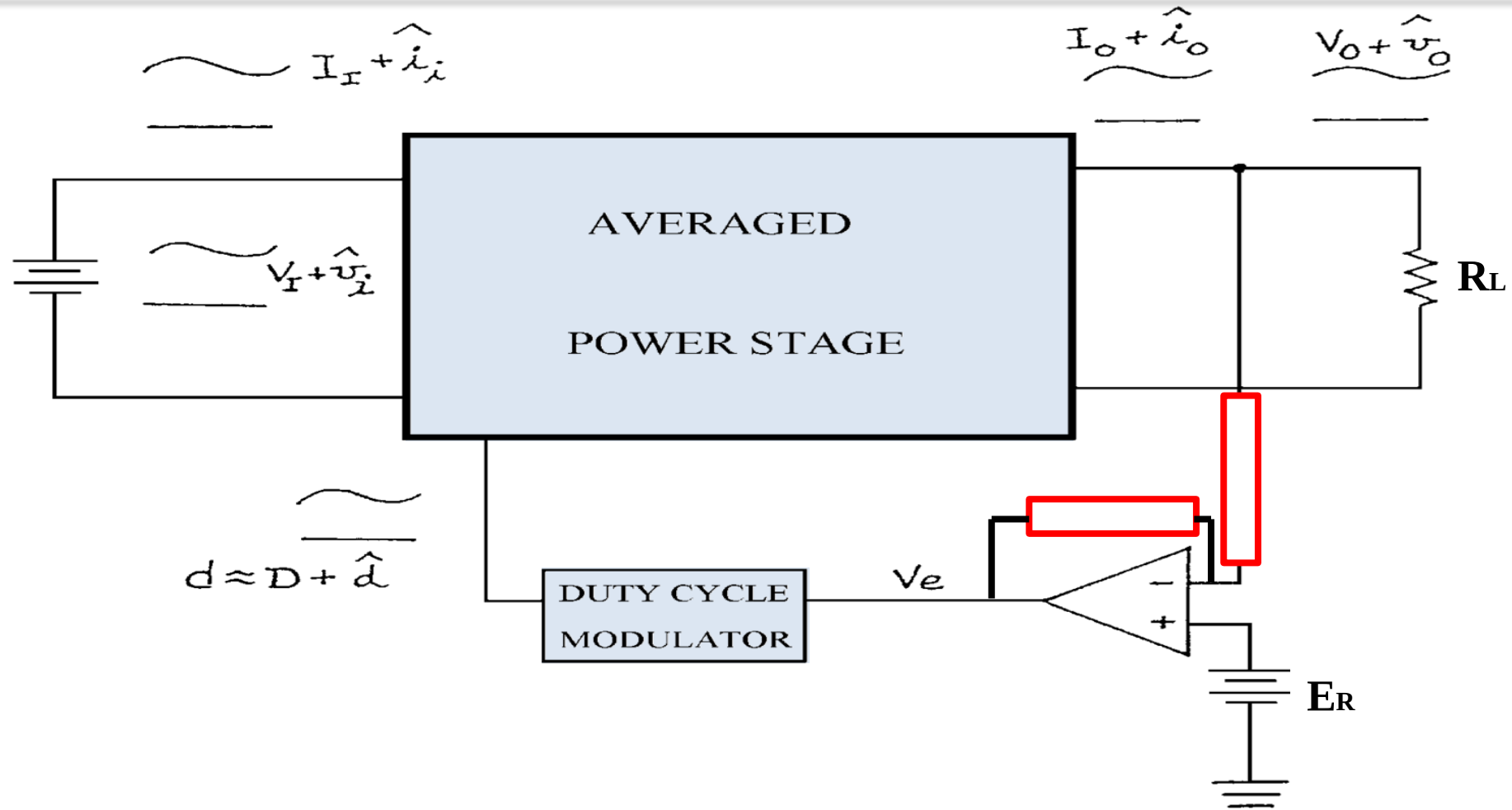
(CONTINUOUS CONDUCTION MODE)

Switching Regulators

Low pass filter (LPF) 1

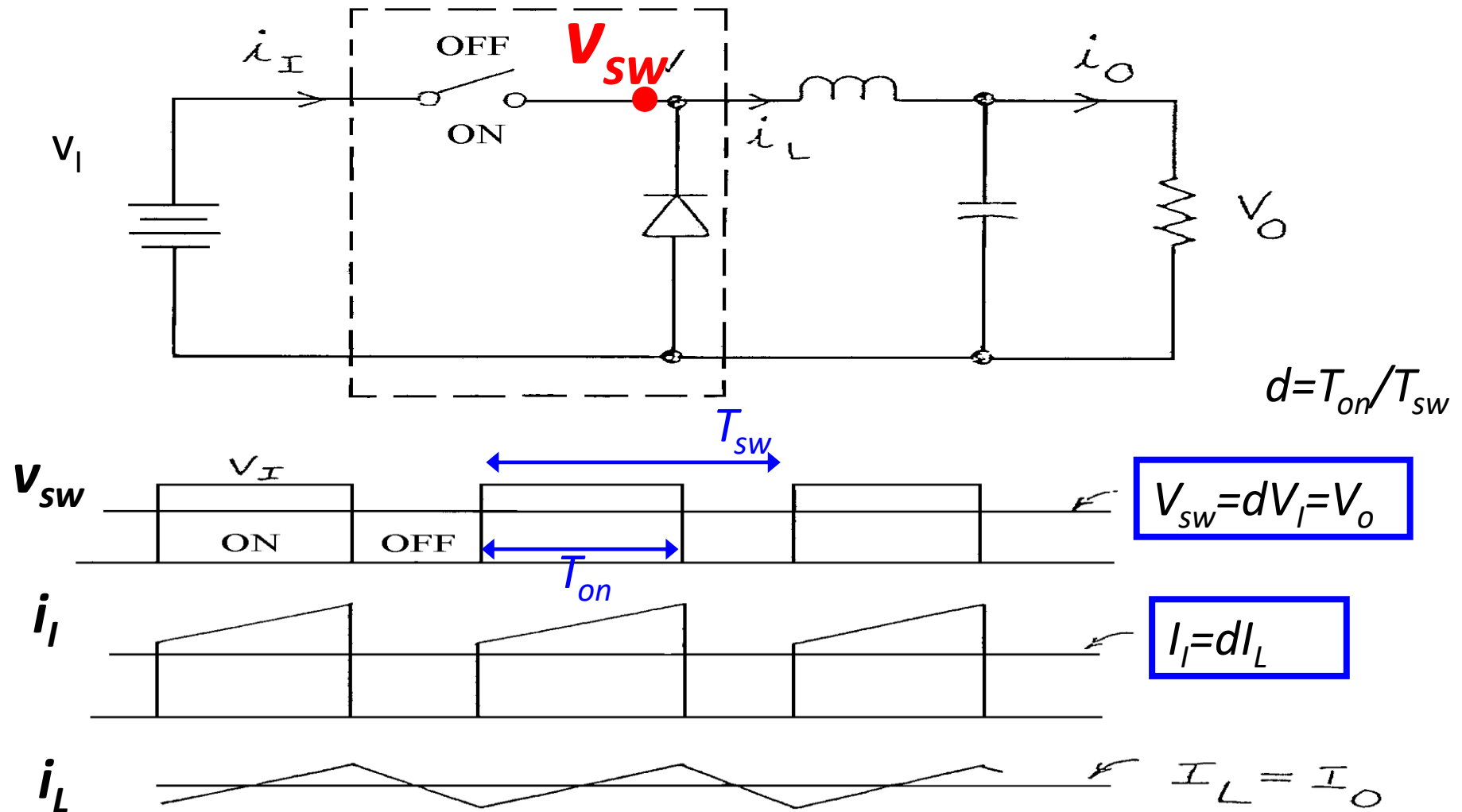


Average Model [1]

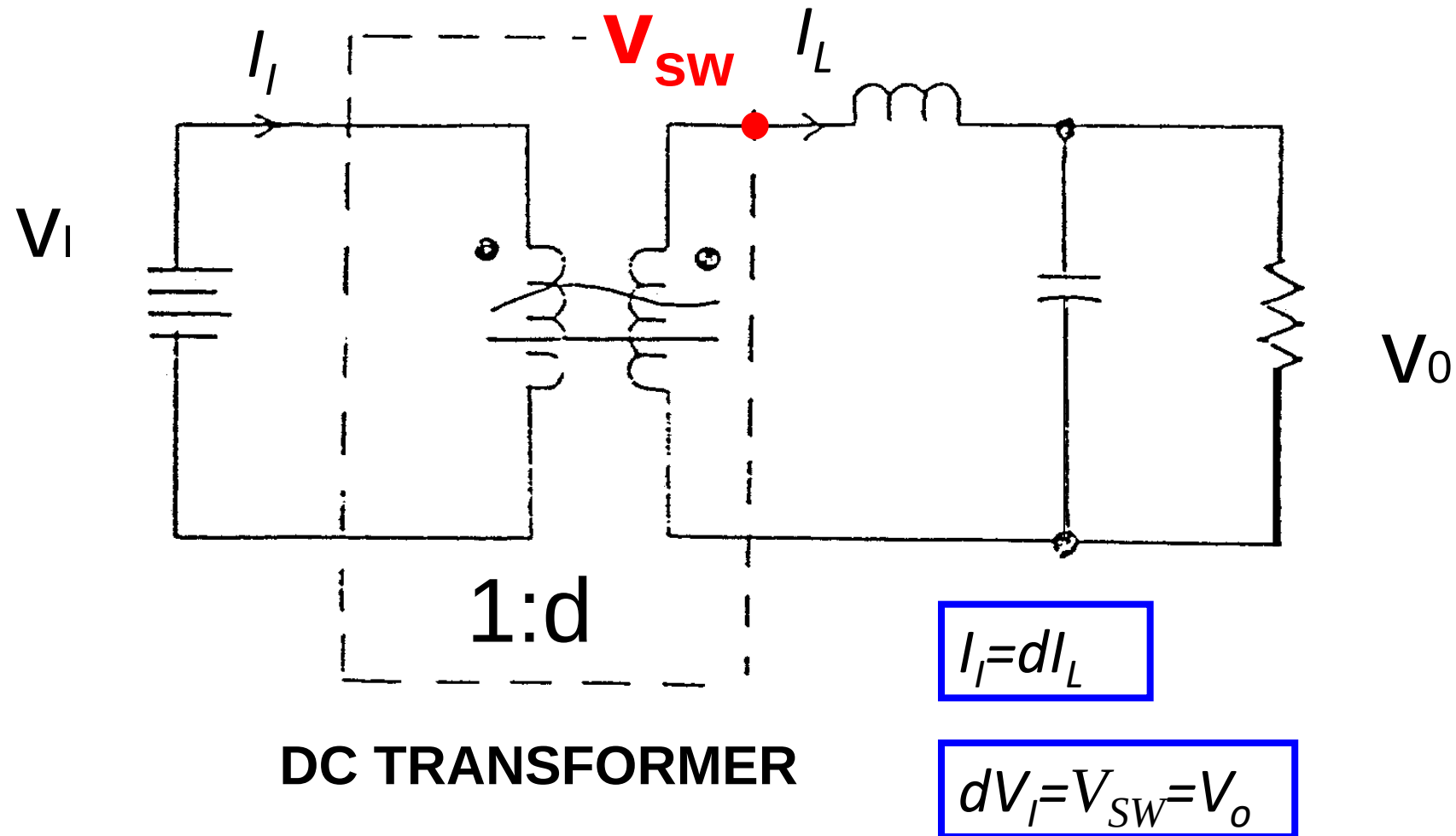


Power Stage Modeling

BUCK CONVERTER

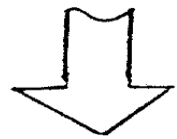
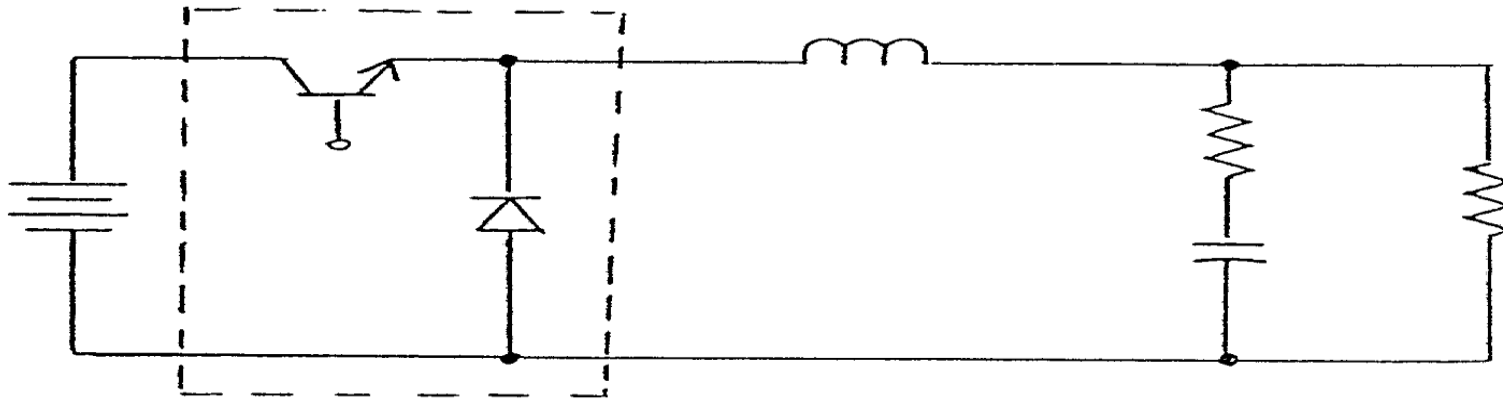


Average Model for Buck Converter

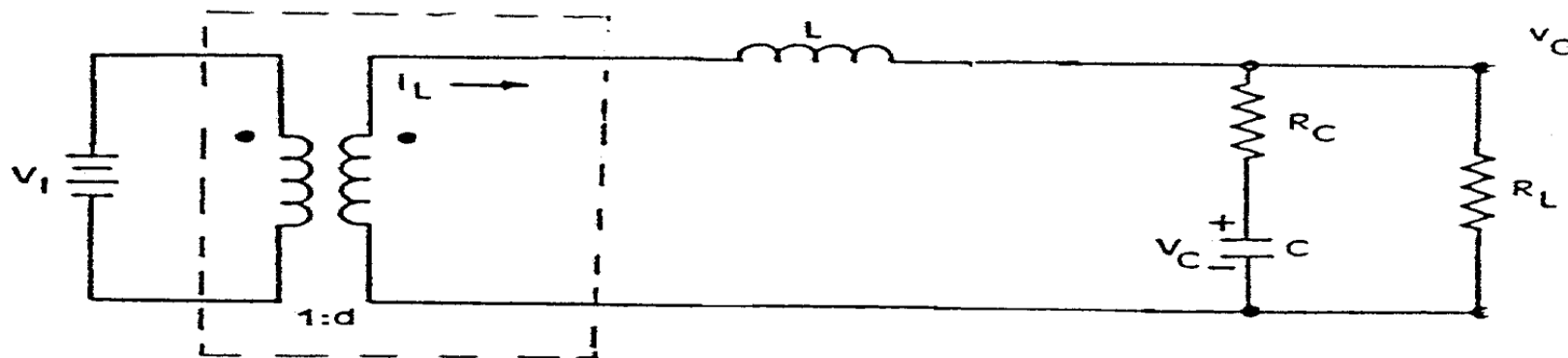


BUCK CONVERTER

Switch network



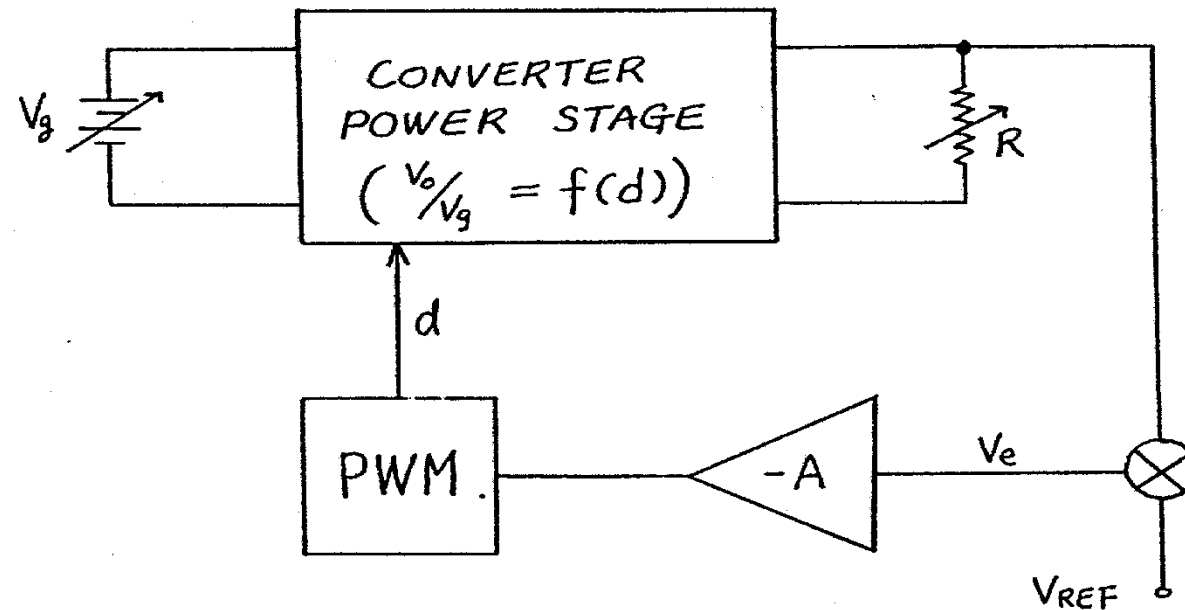
Average Model



Chapter 1.3

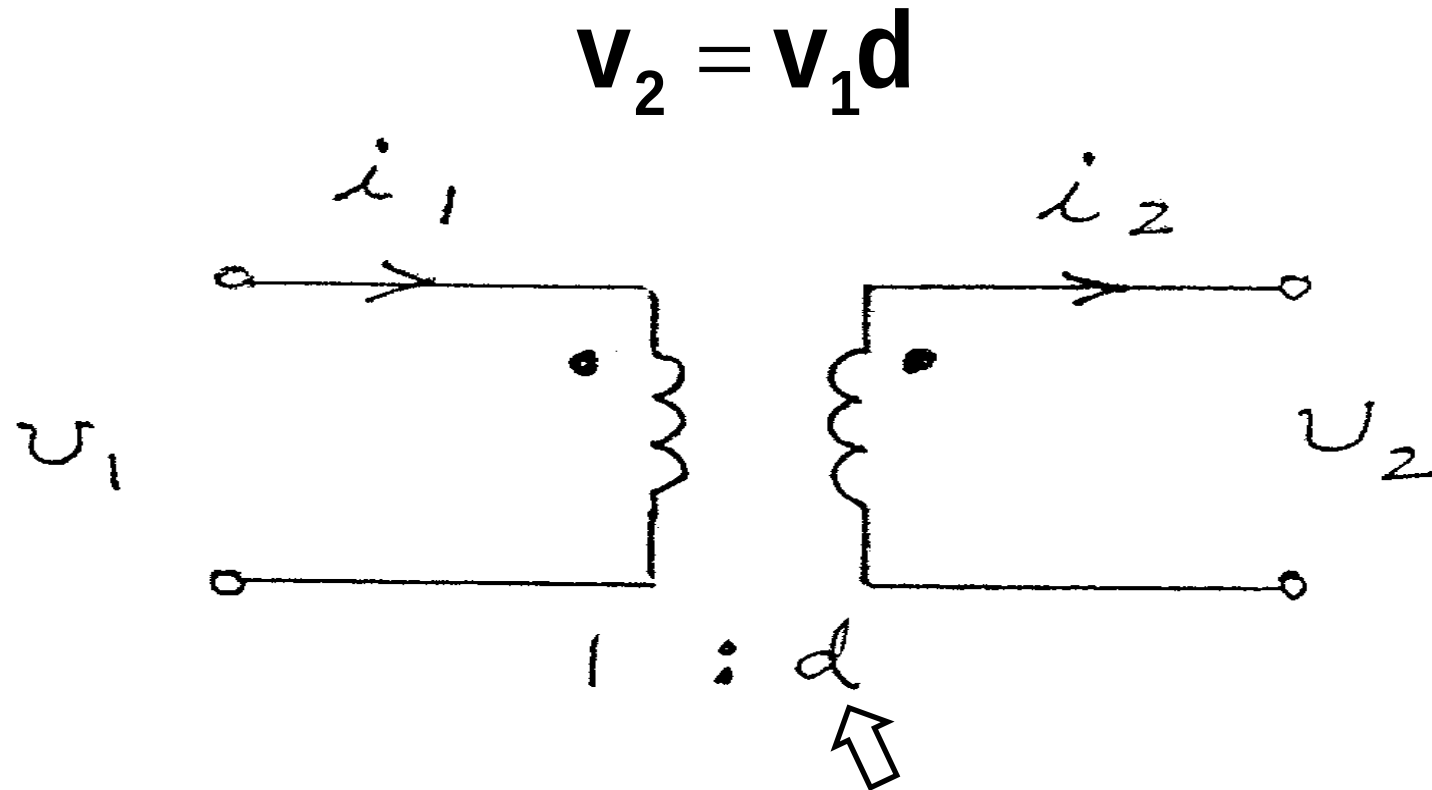
Small Signal Model for Buck Converter

Motivation of Small Signal Modeling



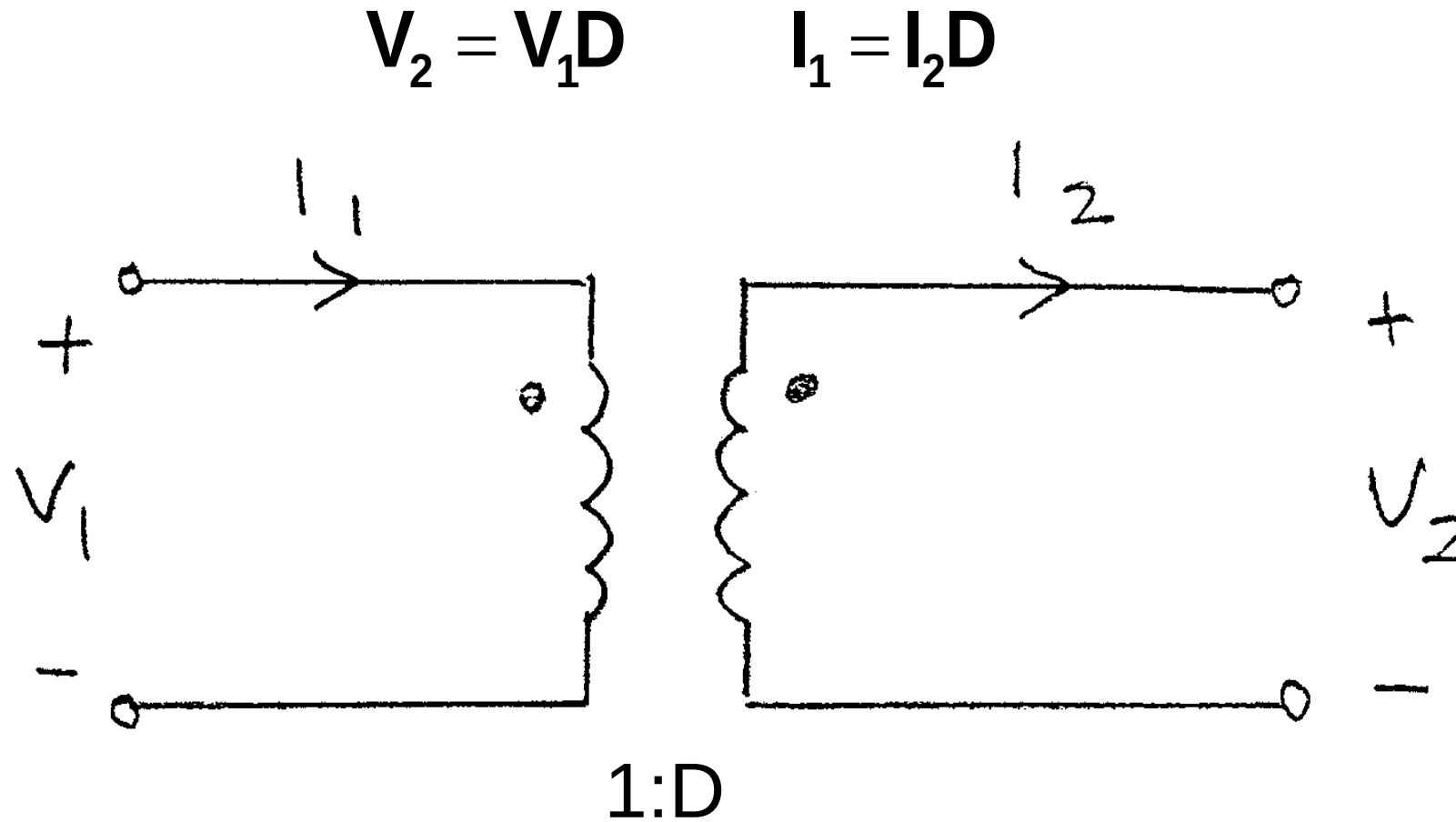
- Determining the response of the fundamental component about a particular dc operating point results in a “*small-signal*” analysis.
- Important in the design of the feedback loop in regulated converters.

1. Average Model of Switch Network



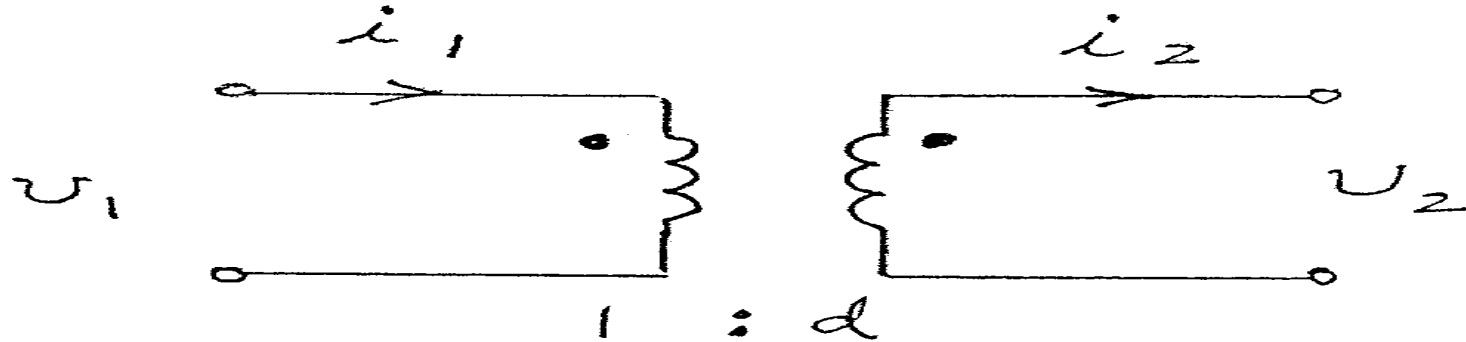
Transformer with variable turns ratio

2. DC Model of Switch Network



3. Add Perturbation

$$\mathbf{v}_1 = \mathbf{V}_1 + \hat{\mathbf{v}}_1 \quad \mathbf{v}_2 = \mathbf{V}_2 + \hat{\mathbf{v}}_2 \quad \mathbf{i}_1 = \mathbf{I}_1 + \hat{\mathbf{i}}_1 \quad \mathbf{i}_2 = \mathbf{I}_2 + \hat{\mathbf{i}}_2 \quad \mathbf{d} = \mathbf{D} + \hat{\mathbf{d}}$$



$$\therefore \mathbf{v}_2 = \mathbf{d} \cdot \mathbf{v}_1 = (\mathbf{D} + \hat{\mathbf{d}})(\mathbf{V}_1 + \hat{\mathbf{v}}_1)$$

$$\mathbf{V}_2 + \hat{\mathbf{v}}_2 = \mathbf{D}\mathbf{V}_1 + \mathbf{D}\hat{\mathbf{v}}_1 + \mathbf{V}_1\hat{\mathbf{d}} + \hat{\mathbf{d}}\hat{\mathbf{v}}_1$$

$$\therefore \mathbf{i}_1 = \mathbf{d} \cdot \mathbf{i}_2 = (\mathbf{D} + \hat{\mathbf{d}})(\mathbf{I}_2 + \hat{\mathbf{i}}_2)$$

$$\mathbf{I}_1 + \hat{\mathbf{i}}_1 = \mathbf{D}\mathbf{I}_2 + \mathbf{D}\hat{\mathbf{i}}_2 + \mathbf{I}_2\hat{\mathbf{d}} + \hat{\mathbf{d}}\hat{\mathbf{i}}_2$$

4. Small-Signal Linearization

$$\hat{d} \ll D$$

$$\hat{v}_1 \ll V_1$$

$$\hat{i}_1 \ll I_1$$

$$\hat{v}_2 \ll V_2$$

$$\hat{i}_2 \ll I_2$$

$$\therefore \underline{V_2} + \hat{v}_2 \approx \underline{DV_1} + D\hat{v}_1 + V_1\hat{d} + \cancel{\hat{d}\hat{v}_1}$$

$$\underline{I_1} + \hat{i}_1 \approx \underline{DI_2} + D\hat{i}_2 + I_2\hat{d} + \cancel{\hat{d}\hat{i}_2}$$

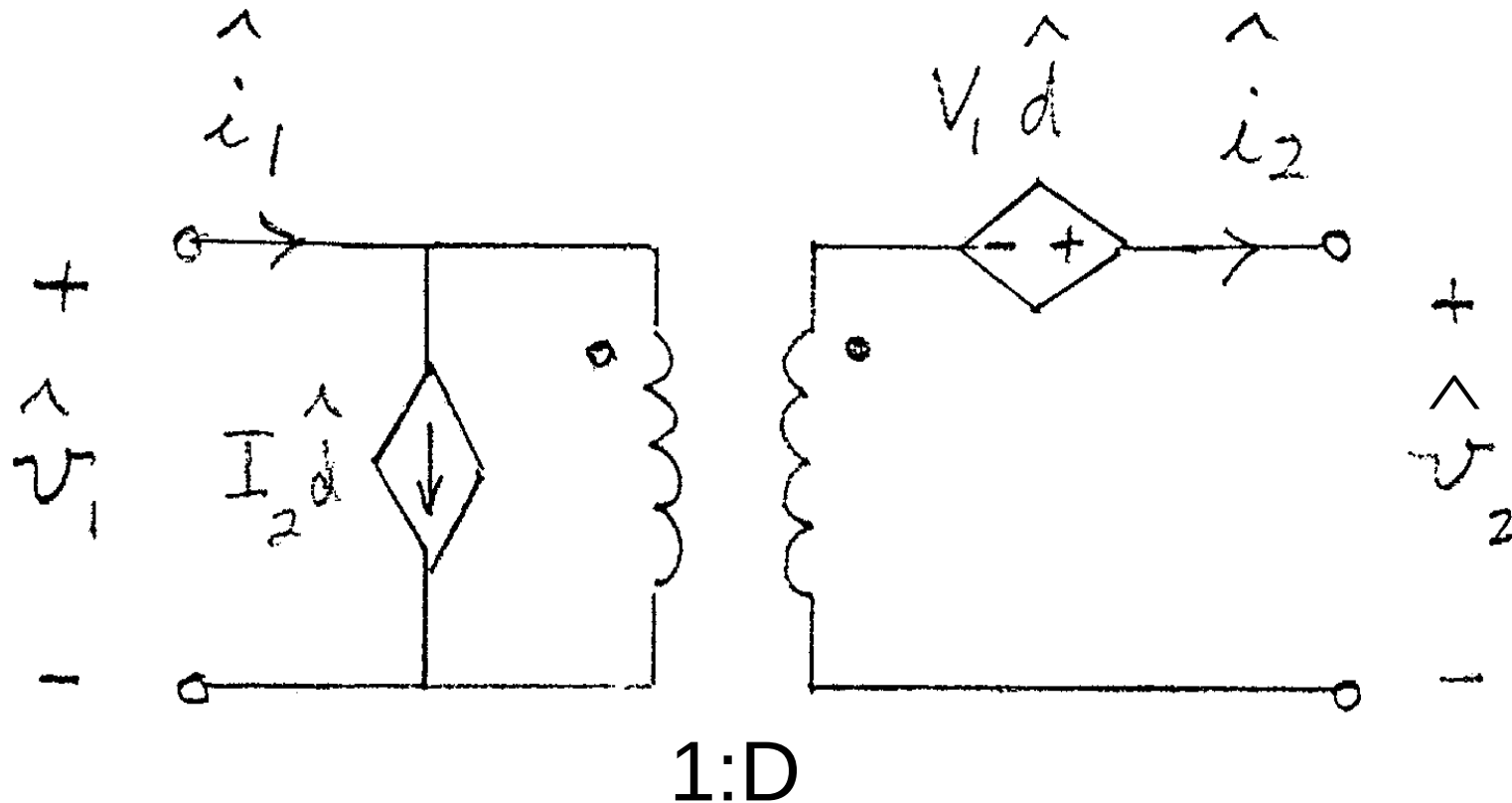
DC operating point

Small-signal linearization

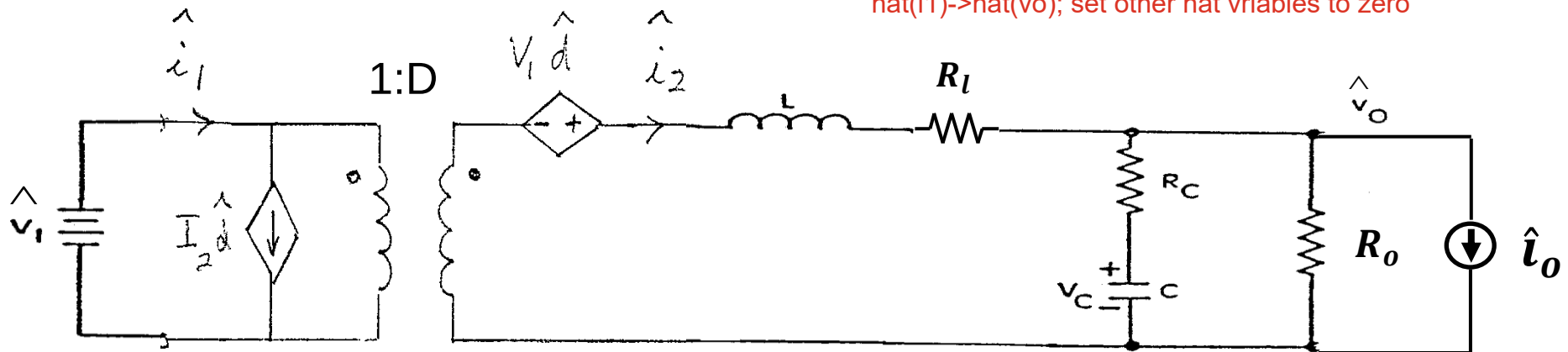
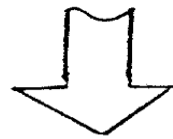
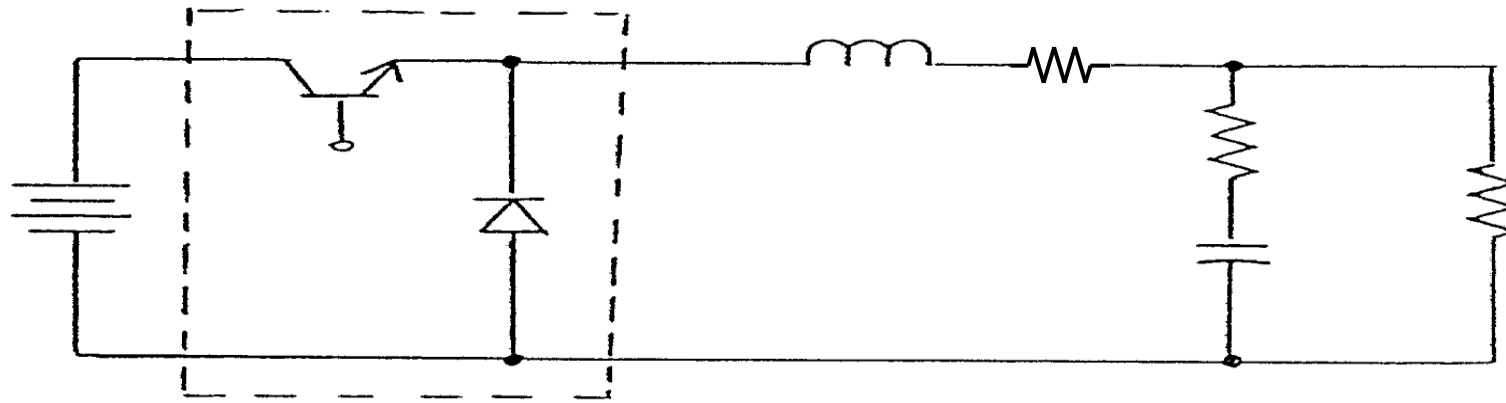
5. Small-Signal Model

$$\hat{\mathbf{v}}_2 \approx \mathbf{D}\hat{\mathbf{v}}_1 + \mathbf{V}_1\hat{\mathbf{d}}$$

$$\hat{\mathbf{i}}_1 \approx \mathbf{D}\hat{\mathbf{i}}_2 + \mathbf{I}_2\hat{\mathbf{d}}$$



Complete Small-Signal Model for Buck



Averaged **small-signal** power stage is a **continuous, linear** system.

Transfer Function of Buck Converter

Transfer Function	Transfer Function
$\frac{\hat{v}_o}{\hat{d}} = V_I \frac{R_o}{R_o + R_l} \frac{1 + sR_c C}{\Delta(s)}$	$\frac{\hat{i}_L}{\hat{d}} = \frac{V_I}{R_o + R_l} \frac{1 + sC(R_c + R_l)}{\Delta(s)}$
$\frac{\hat{v}_o}{\hat{v}_I} = D \frac{R_o}{R_o + R_l} \frac{1 + sR_c C}{\Delta(s)}$ Audio susceptibility	$\frac{\hat{i}_L}{\hat{v}_I} = \frac{D}{R_o + R_l} \frac{1 + sC(R_c + R_l)}{\Delta(s)}$
$\frac{\hat{v}_o}{\hat{i}_o} = - \frac{R_o R_l}{R_o + R_l} \frac{(1 + sR_c C)(1 + sL/R_l)}{\Delta(s)}$ output impedance	$\frac{\hat{i}_L}{\hat{i}_o} = \frac{R_o}{R_o + R_l} \frac{1 + sR_c C}{\Delta(s)}$

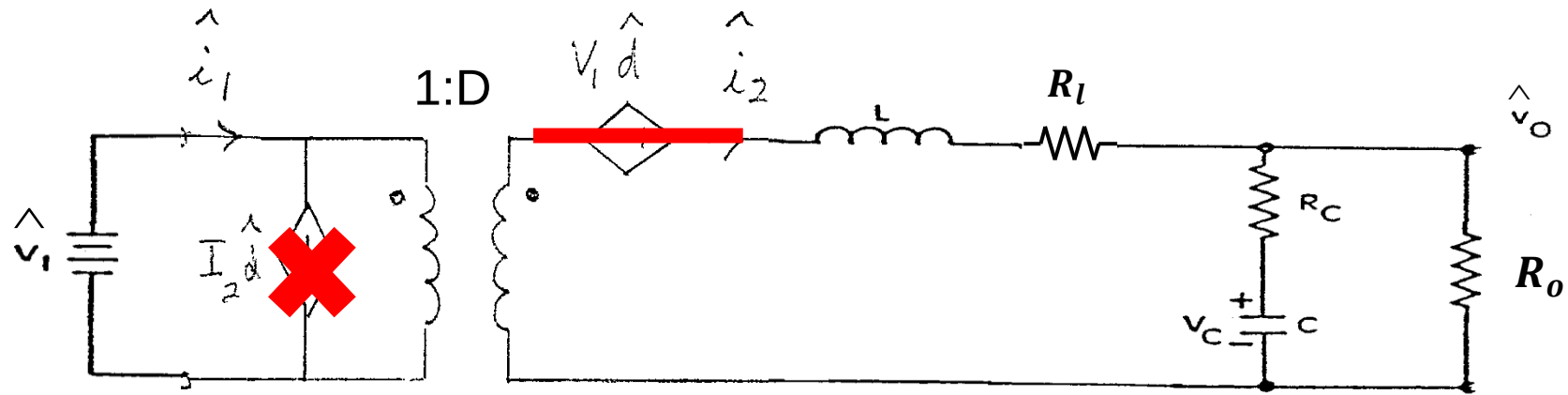
zero comes from ESR of capacitor zero comes from DCR of inductor

small impedance means v_o barely change with i_o change, good for CV mode dc/dc converter

$$\Delta(s) = 1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2} \quad \omega_0 = \frac{1}{\sqrt{LC \frac{R_o + R_c}{R_o + R_l}}} \approx \frac{1}{\sqrt{LC}} \quad Q = \frac{1}{\omega_0} \frac{1}{\frac{L}{R_o + R_l} + C \left(R_c + \frac{R_o R_l}{R_o + R_l} \right)}$$

R_c : ESR of output capacitor; R_l : Inductor DCR; R_o : Load resistor

Open-Loop Line-to-Output Transfer Function



$$\frac{\hat{v}_O}{\hat{v}_I} = D \frac{R_o // (R_c + 1/sC)}{(sL + R_l) + R_o // (R_c + 1/sC)} = D \frac{R_o}{R_o + R_l} \frac{1 + sR_c C}{\Delta(s)} \approx D \frac{1 + sR_c C}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$$

$$\omega_0 = \frac{1}{\sqrt{LC \frac{R_o + R_c}{R_o + R_l}}} \approx \frac{1}{\sqrt{LC}}$$

$$Q = \frac{1}{\omega_0} \frac{1}{\frac{L}{R_o + R_l} + C \left(R_c + \frac{R_o R_l}{R_o + R_l} \right)} \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{R_o} + C(R_c + R_l)}$$

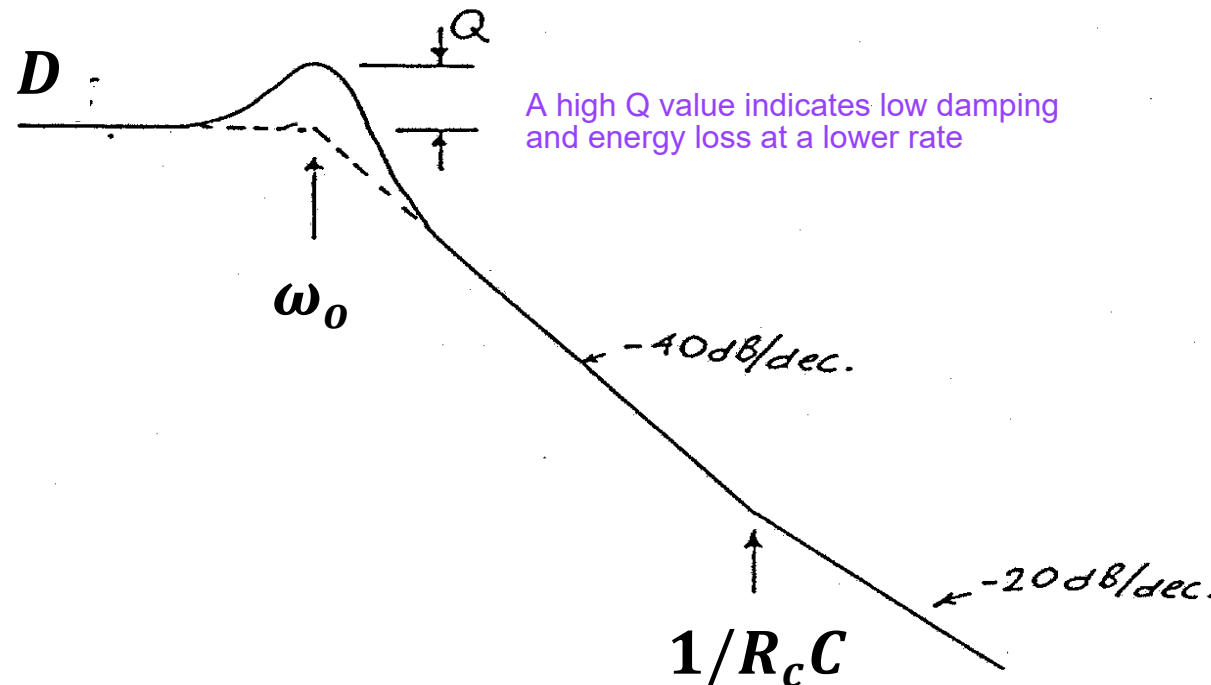
Open-Loop Line-to-Output Transfer Function

The asymptotic magnitude Bode plot is shown below:

$$\frac{\hat{v}_o}{\hat{v}_I} \approx D \frac{1 + sR_c C}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$$

$$\omega_0 \approx \frac{1}{\sqrt{LC}}$$

$$Q \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{R_o} + C(R_c + R_l)}$$



Input Voltage Step Response

$$F_s = 300\text{kHz}$$

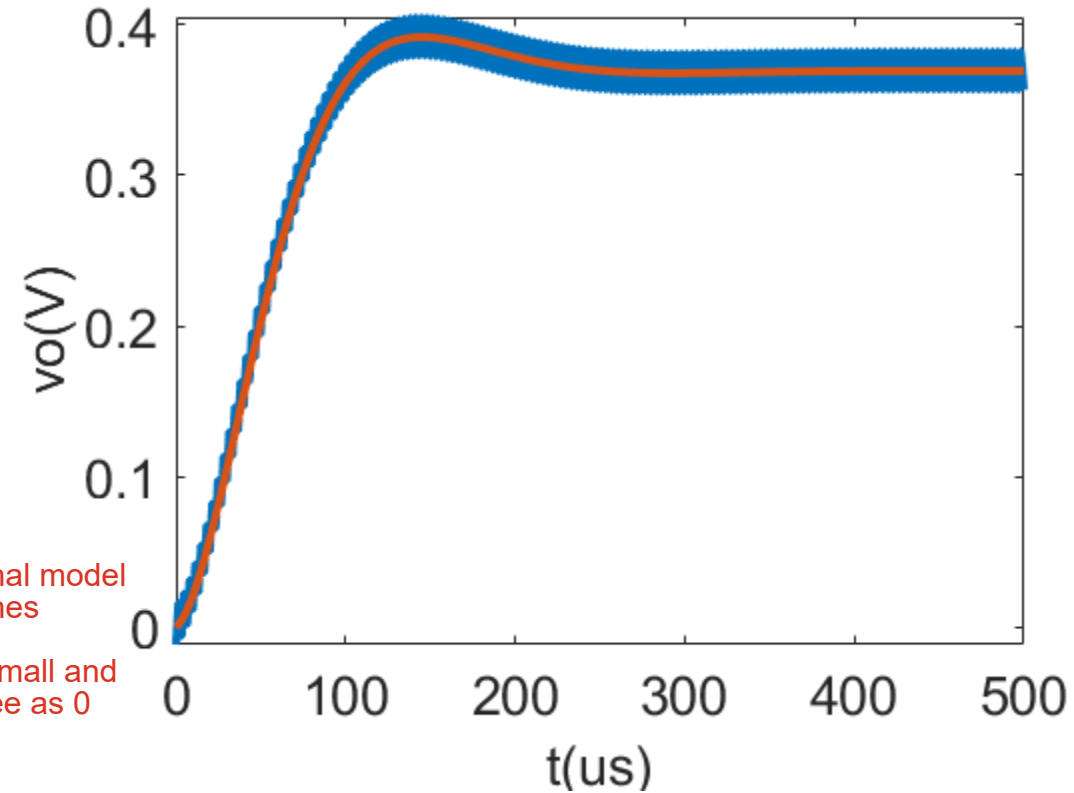
$$T_s = 3.3\mu\text{s}$$

$v_o - V_o$ from switching circuit

v_o from transfer function

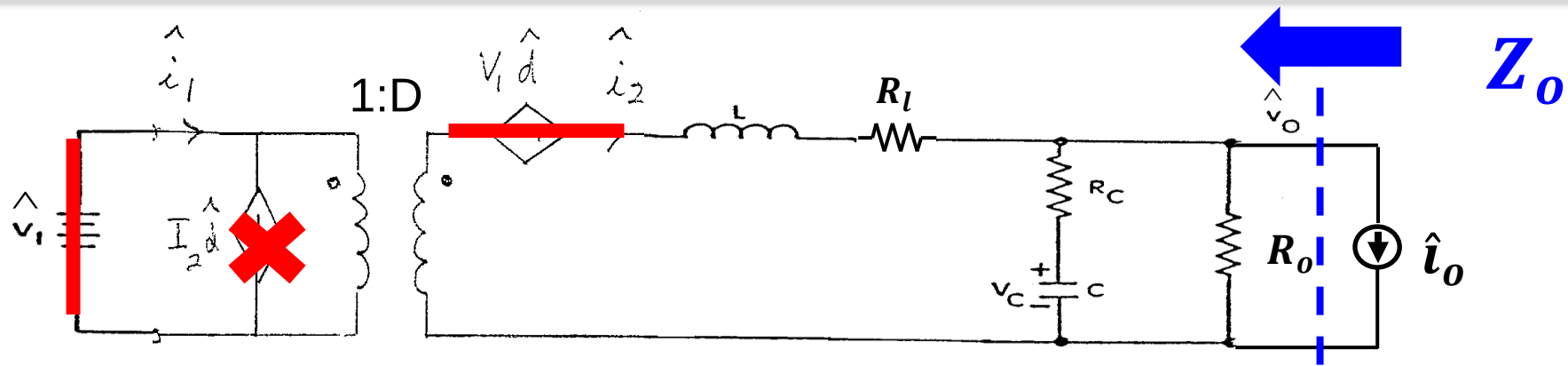
the duty cycle is fixed in open loop system, small signal model will same to large signal model, so step up test matches

In buck converter, the output impedance is same in small and large signal model as the source impedance can be seen as 0



The transfer function is a good approximation of the system.

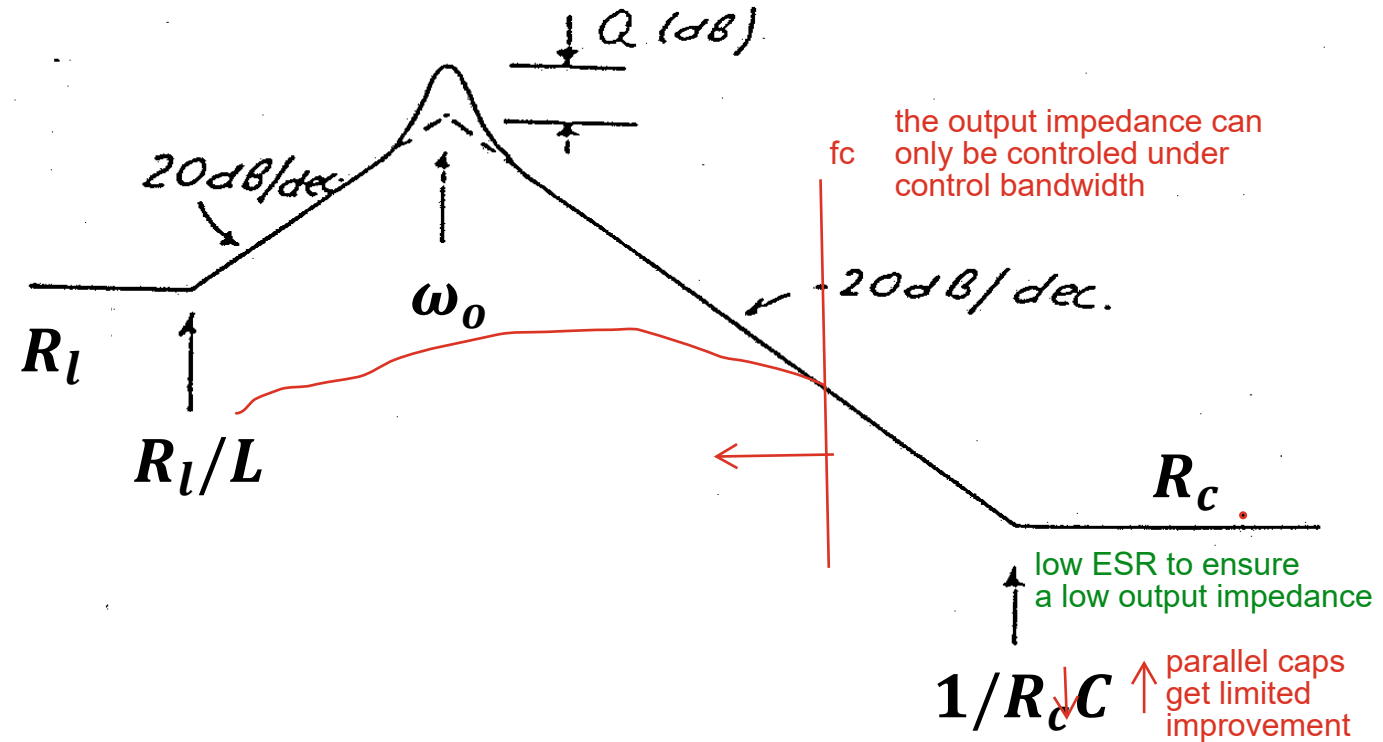
Open-Loop Output Impedance



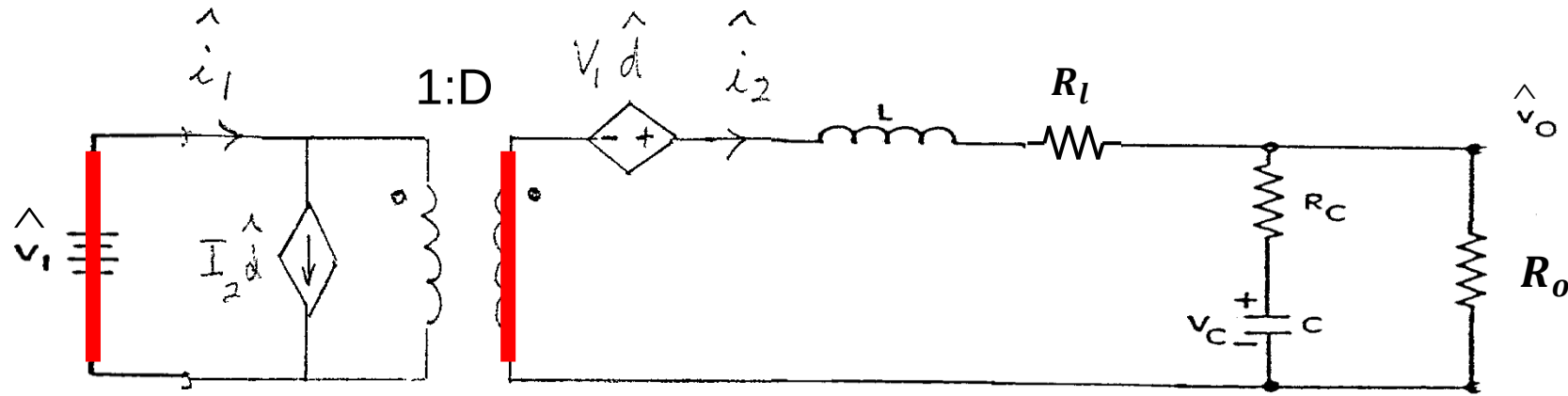
$$Z_o = -\frac{\hat{v}_o}{\hat{i}_o} \approx \frac{R_o R_l}{R_o + R_l} \frac{(1 + s R_c C)(1 + s L / R_l)}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$$

$$\omega_0 \approx \frac{1}{\sqrt{LC}}$$

$$Q \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{R_o} + C(R_c + R_l)}$$



Open-Loop Duty-Cycle-to-Output Transfer Function



$$\frac{\hat{v}_o}{\hat{d}} \approx V_I \frac{1 + sR_c C}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$$

$$\omega_0 \approx \frac{1}{\sqrt{LC}}$$

$$Q \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{R_o} + C(R_c + R_l)}$$

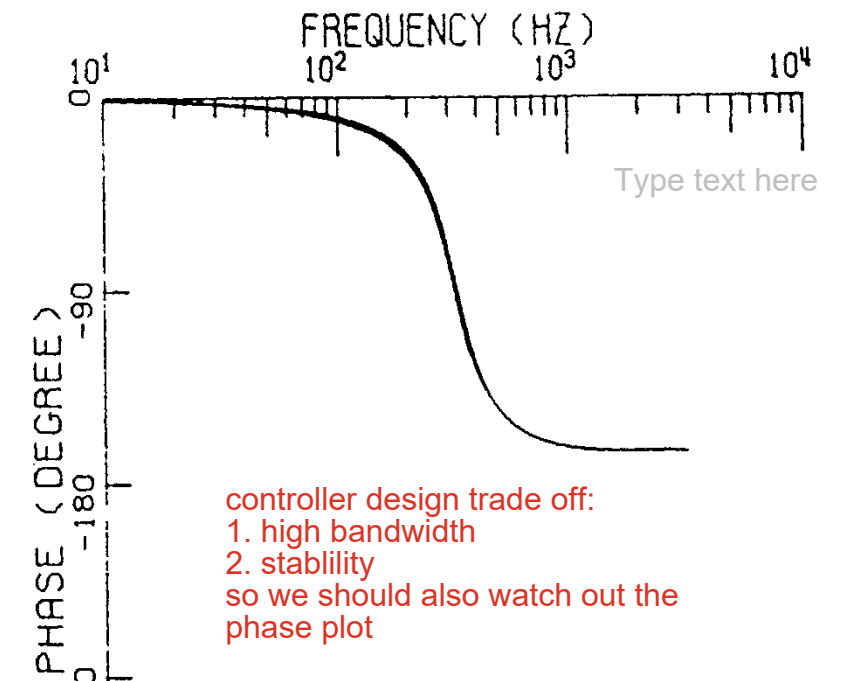
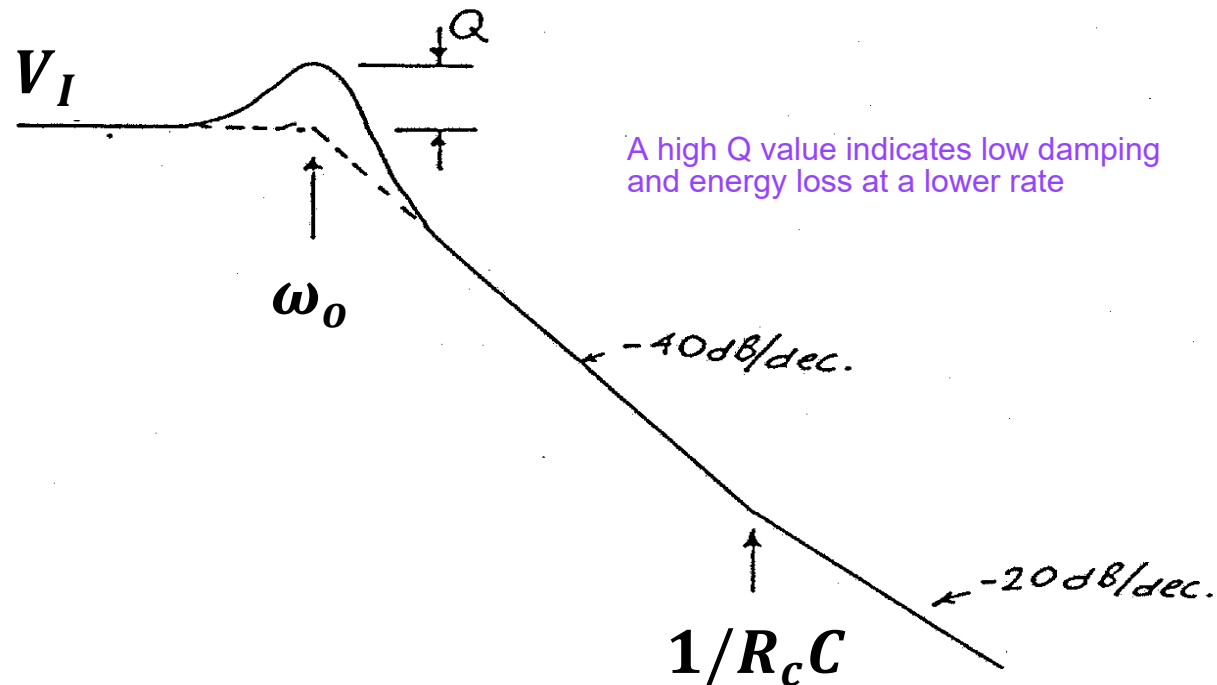
Open-Loop Duty-Cycle-to-Output Transfer Function

The asymptotic magnitude Bode plot is shown below:

$$\frac{\hat{v}_o}{\hat{d}} \approx V_I \frac{1 + sR_c C}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$$

$$\omega_0 \approx \frac{1}{\sqrt{LC}}$$

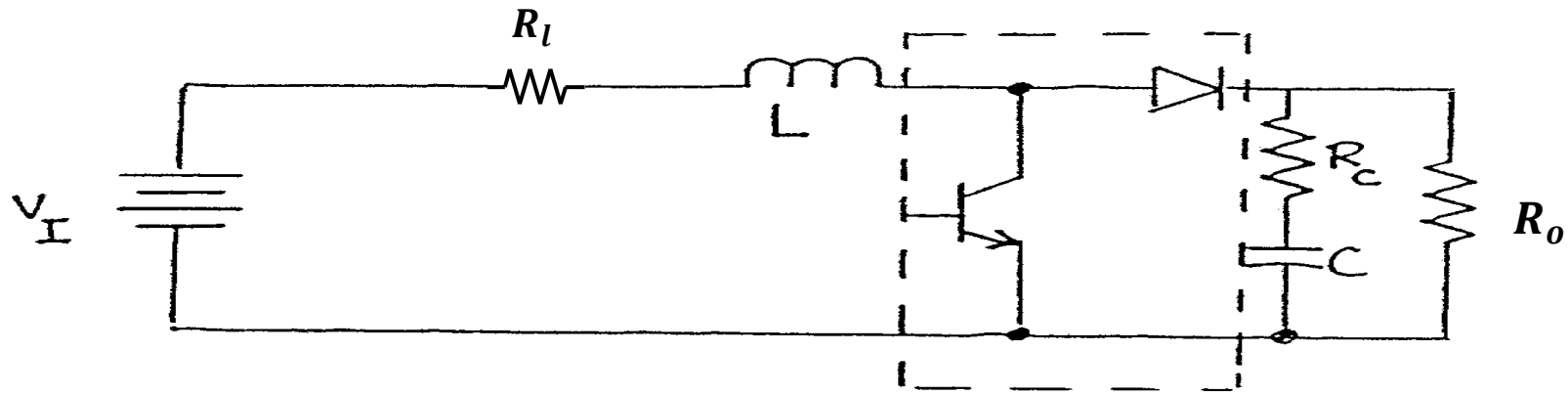
$$Q \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{R_o} + C(R_c + R_l)}$$



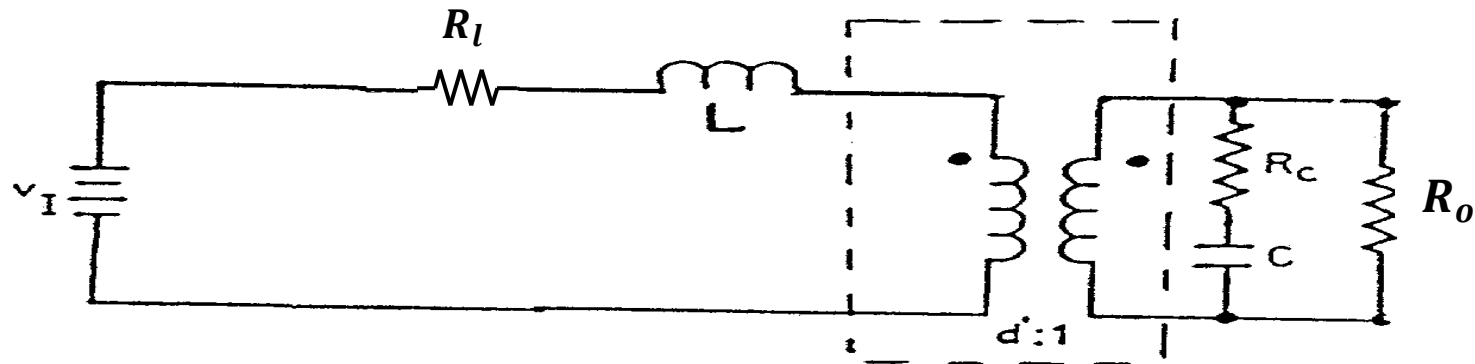
Chapter 1.4

Small Signal Model for Boost Converter

BOOST CONVERTER

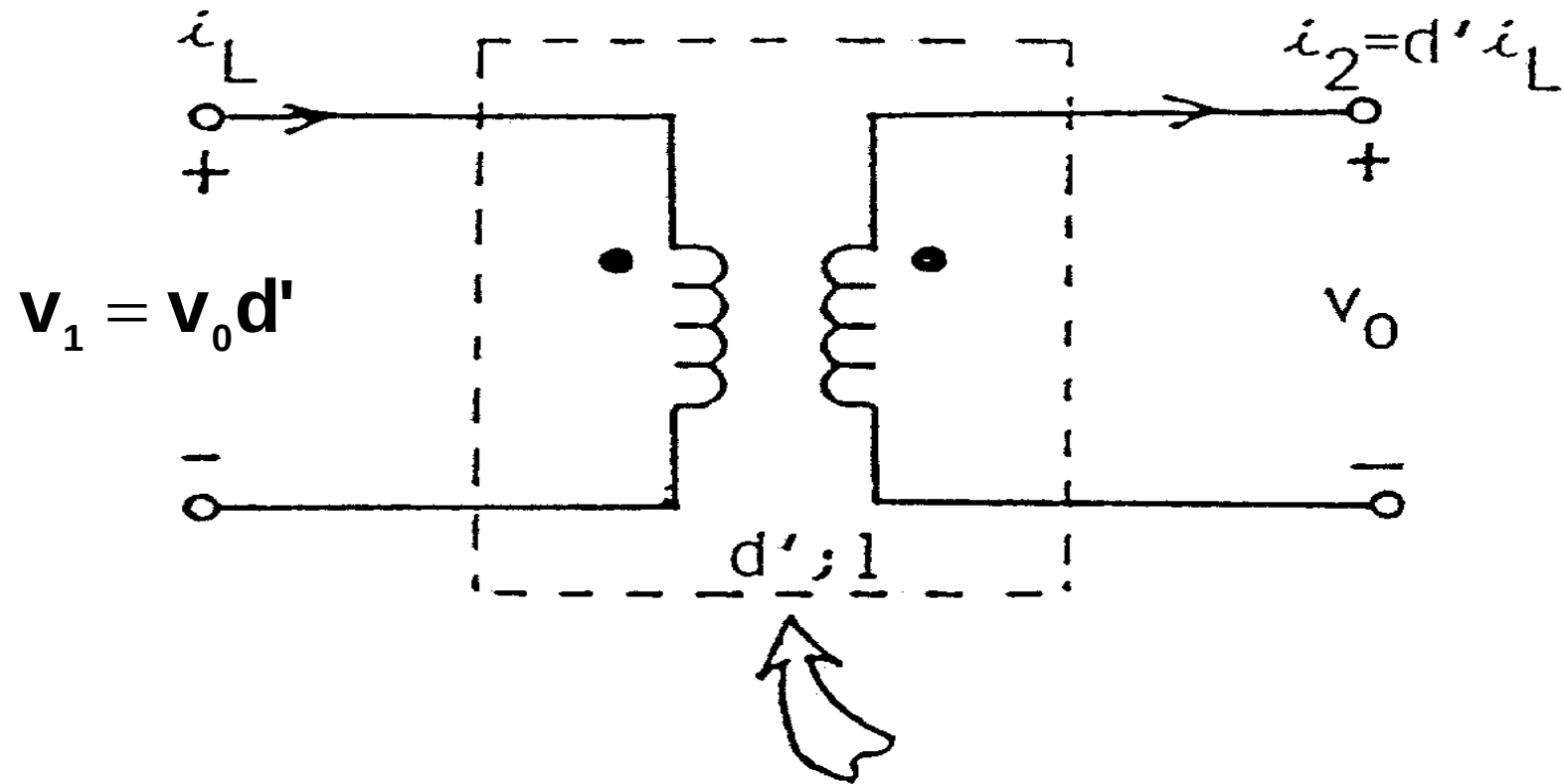


↓ AVERAGE



$$d' \triangleq 1 - d$$

Average Model



Transformer with variable turns ratio

Add Perturbation

$$\mathbf{d}' = \mathbf{1} - (\mathbf{D} + \hat{\mathbf{d}}) = \mathbf{D}' - \hat{\mathbf{d}}$$

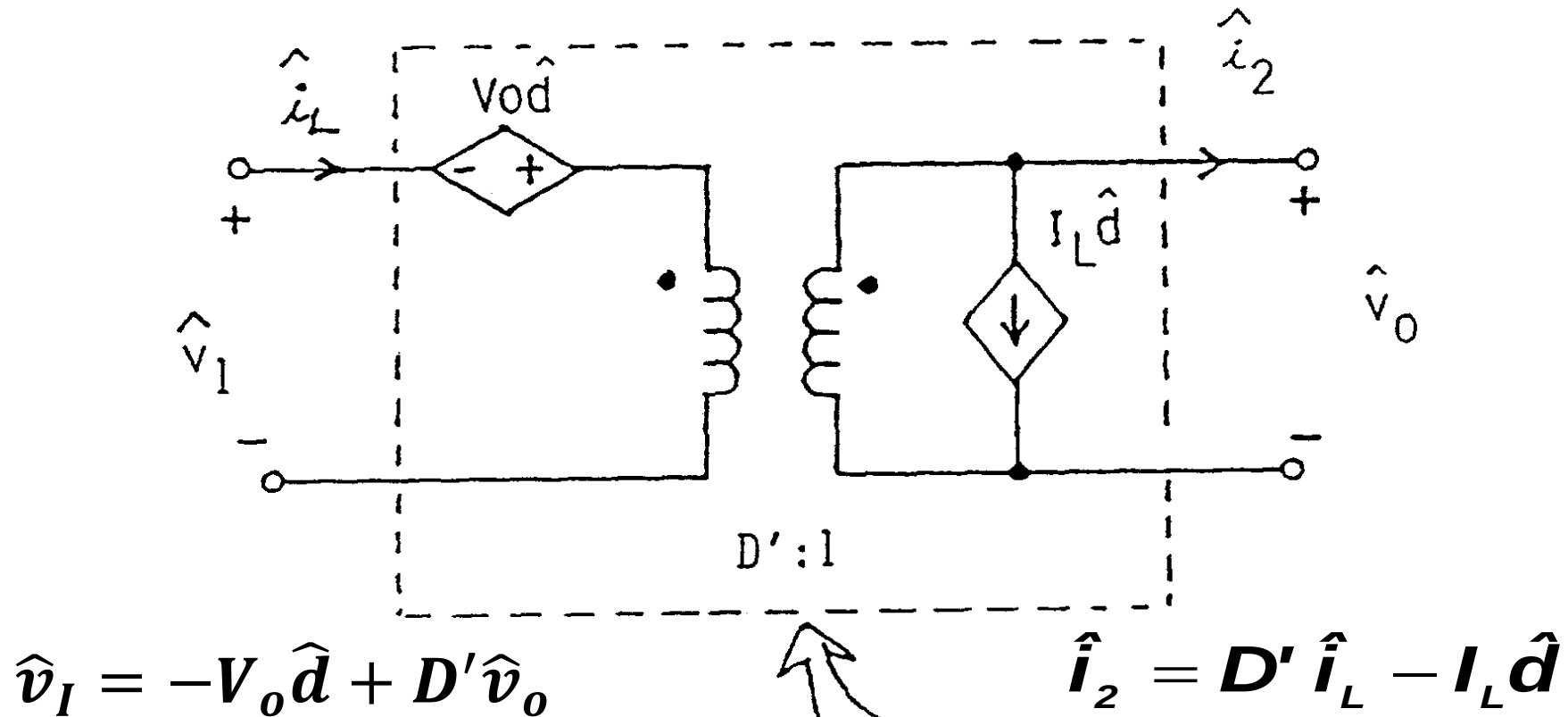
$$\mathbf{i}_L = \mathbf{I}_L + \hat{\mathbf{i}}_L$$

$$\mathbf{v}_1 = (\mathbf{D}' - \hat{\mathbf{d}})(\mathbf{v}_o + \hat{\mathbf{v}}_o)$$

$$\mathbf{v}_o = \mathbf{V}_o + \hat{\mathbf{v}}_o$$

$$\mathbf{i}_2 = (\mathbf{D}' - \hat{\mathbf{d}})(\mathbf{I}_L + \hat{\mathbf{i}}_L)$$

Small Signal Model



Transformer with
constant turns ratio



Transfer Function of Boost Converter

Transfer Function	Transfer Function
$\frac{\hat{v}_o}{\hat{d}} = \frac{V_I}{(D')^2} \frac{R_o}{R_o + R_{le}} \frac{(1 + sR_c C)(1 - s/\omega_a)}{\Delta(s)}$	$\frac{\hat{i}_L}{\hat{d}} = \frac{2}{(D')^3} \frac{V_I}{R_o + R_{le}} \frac{1 + sC(R_c + R_o/2)}{\Delta(s)}$
$\frac{\hat{v}_o}{\hat{v}_I} = \frac{1}{D'} \frac{R_o}{R_o + R_{le}} \frac{1 + sR_c C}{\Delta(s)}$	$\frac{\hat{i}_L}{\hat{v}_I} = \frac{1}{(D')^2} \frac{1}{R_o + R_{le}} \frac{1 + sC(R_o + R_c)}{\Delta(s)}$
$\frac{\hat{v}_o}{\hat{i}_o} = - \frac{R_o R_{le}}{R_o + R_{le}} \frac{(1 + sR_c C)(1 + sL_e/R_{le})}{\Delta(s)}$	$\frac{\hat{i}_L}{\hat{i}_o} = \frac{1}{D'} \frac{R_o}{R_o + R_{le}} \frac{1 + sR_c C}{\Delta(s)}$

RHZ is a additional delay

$$\Delta(s) = 1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}$$

duty cycle take part in poles

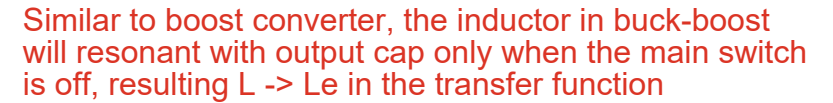
$$L_e = \frac{L}{(D')^2}$$

$$\omega_0 = \frac{1}{\sqrt{L_e C} \frac{R_o + R_c}{R_o + R_{le}}}$$

$$R_{le} = \frac{R_l}{(D')^2}$$

$$Q = \frac{1}{\omega_0} \frac{1}{\frac{L_e}{R_o + R_{le}} + C \left(R_c + \frac{R_o R_{le}}{R_o + R_{le}} \right)}$$

$$\omega_a = \frac{R_o - R_{le}}{L_e} \approx \frac{(D')^2 R_o}{L}$$



$$R_{le} = \frac{R_l}{(D')^2}$$



$$\frac{\hat{v}_o}{\hat{v}_I} = \frac{1}{D'} \frac{R_o}{R_o + R_{le}} \frac{1 + sR_c C}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}} \approx \frac{1}{D'} \frac{1 + sR_c C}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$$

$$\omega_0 = \frac{1}{\sqrt{L_e C \frac{R_o + R_c}{R_o + R_{le}}}} \approx \frac{D'}{\sqrt{LC}}$$

$$Q = \frac{1/\omega_o}{\frac{L_e}{R_o + R_{le}} + C \left(R_c + \frac{R_o R_{le}}{R_o + R_{le}} \right)} \approx \frac{1/\omega_o}{\frac{L}{(D')^2 R_o} + C \left(R_c + \frac{R_l}{(D')^2} \right)}$$

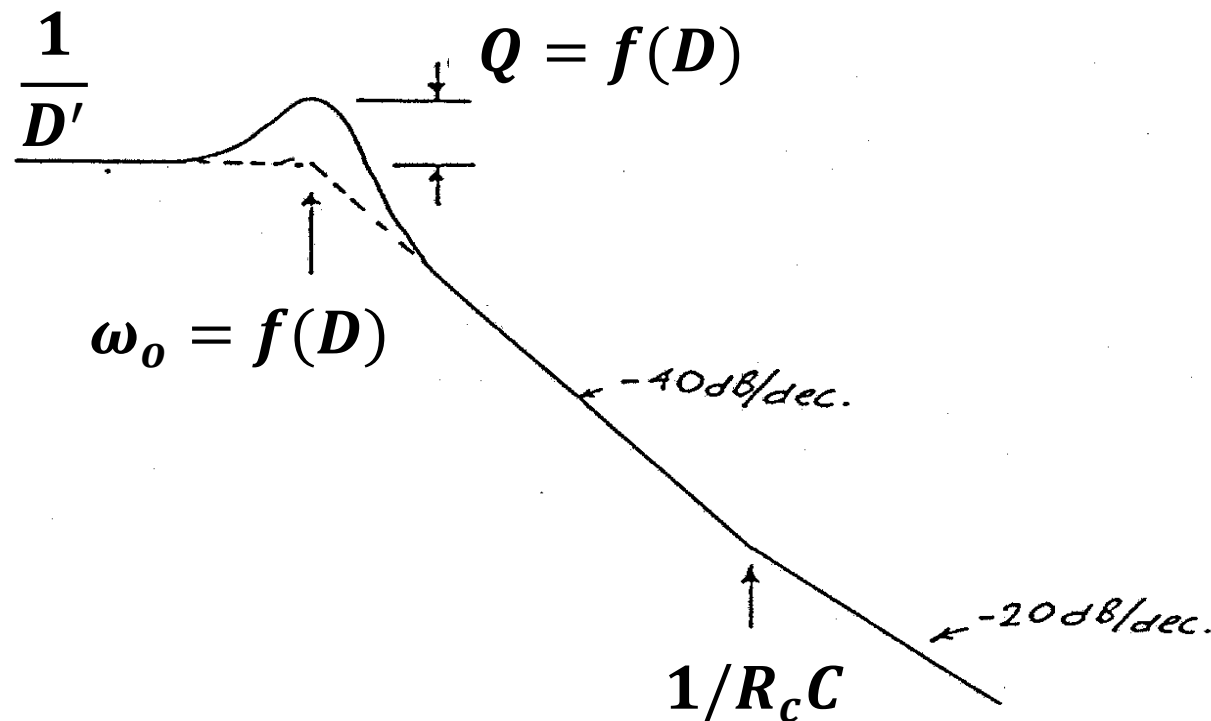
Open-Loop Line-to-Output Transfer Function

The asymptotic magnitude Bode plot is shown below:

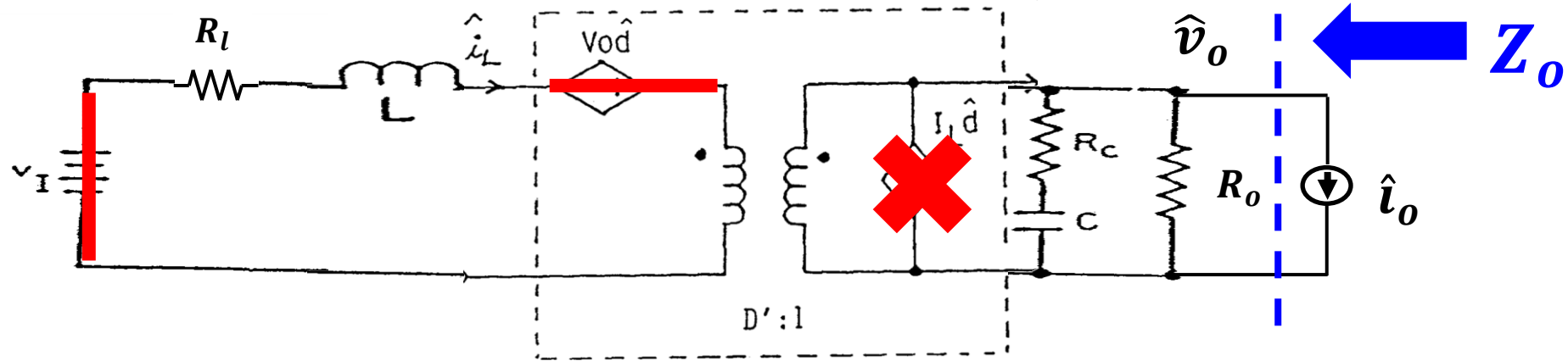
$$\frac{\hat{v}_o}{\hat{d}} \approx \frac{1}{D'} \frac{1 + sR_c C}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$$

$$\omega_0 \approx \frac{D'}{\sqrt{LC}}$$

$$Q \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{(D')^2 R_o} + C \left(R_c + \frac{R_l}{(D')^2} \right)}$$



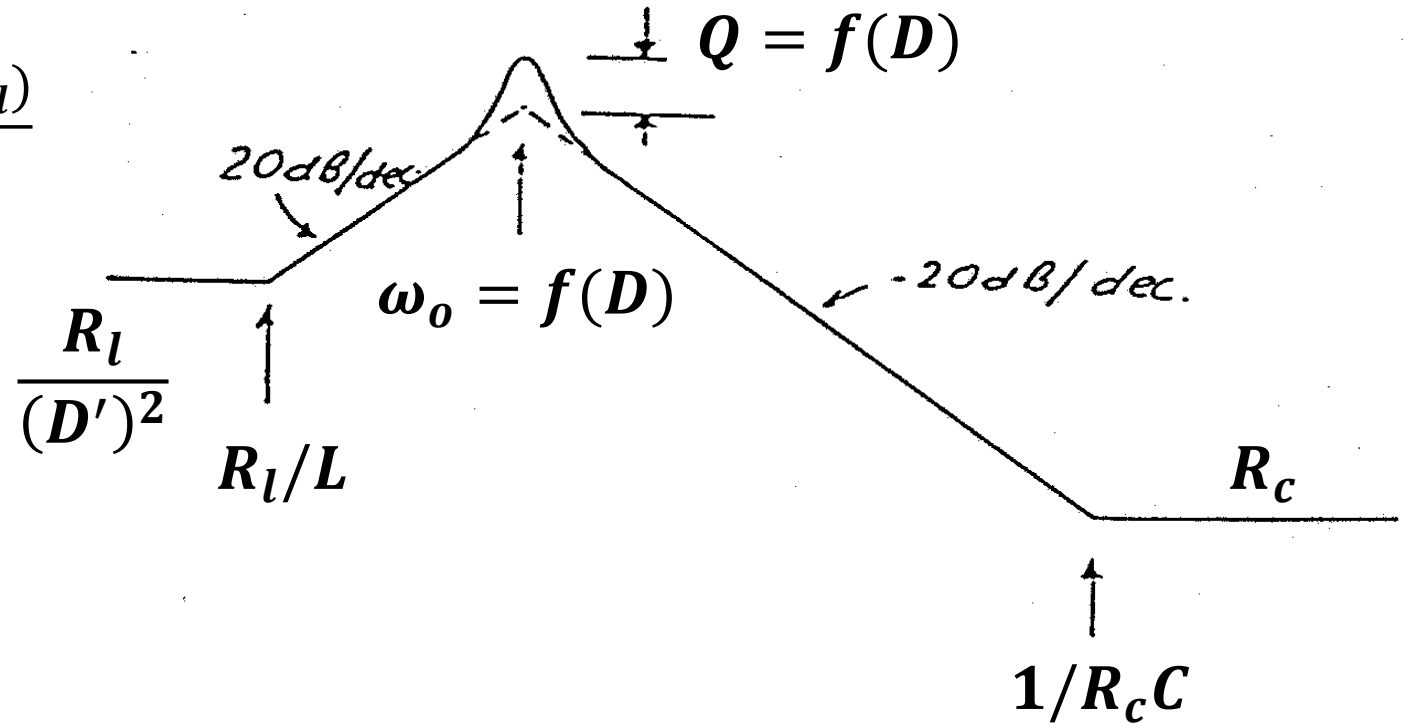
Open-Loop Output Impedance



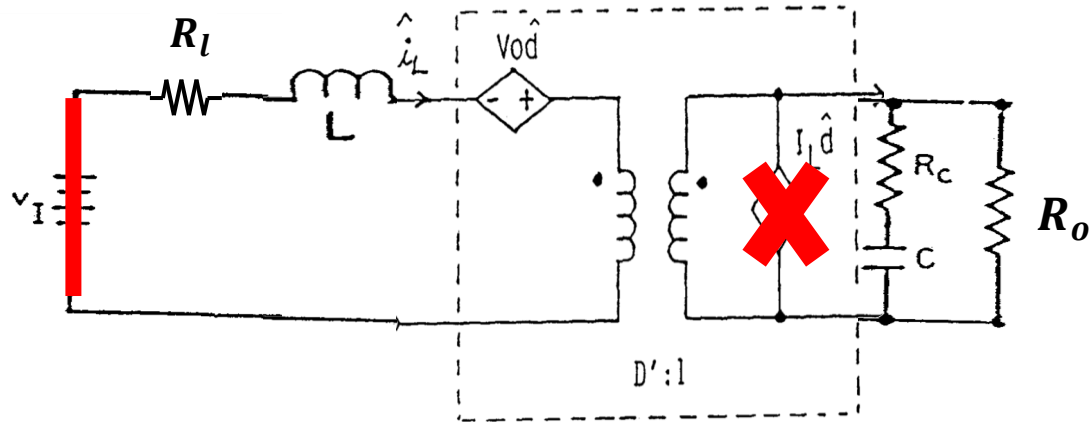
$$Z_o = -\frac{\hat{v}_o}{\hat{i}_o} \approx \frac{R_o R_{le}}{R_o + R_{le}} \frac{(1 + sR_c C)(1 + sL/R_l)}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$$

$$\omega_0 \approx \frac{D'}{\sqrt{LC}}$$

$$Q \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{(D')^2 R_o} + C \left(R_c + \frac{R_l}{(D')^2} \right)}$$

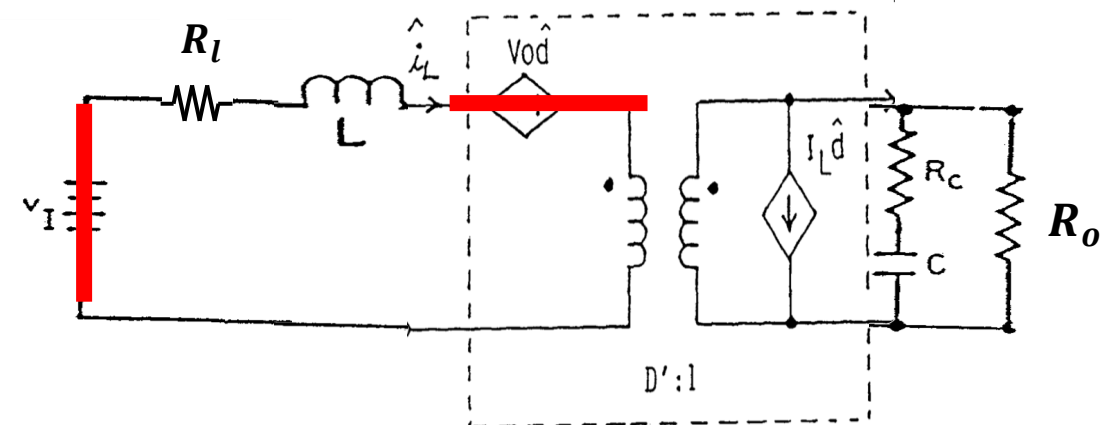


Open-Loop Duty-Cycle-to-Output Transfer Function



From controlled voltage source:

$$\frac{\hat{v}_o}{\hat{d}} = \frac{V_I}{(D')^2} \frac{R_o}{R_o + R_{le}} \frac{1 + sCR_c}{\Delta(s)}$$



From controlled current source:

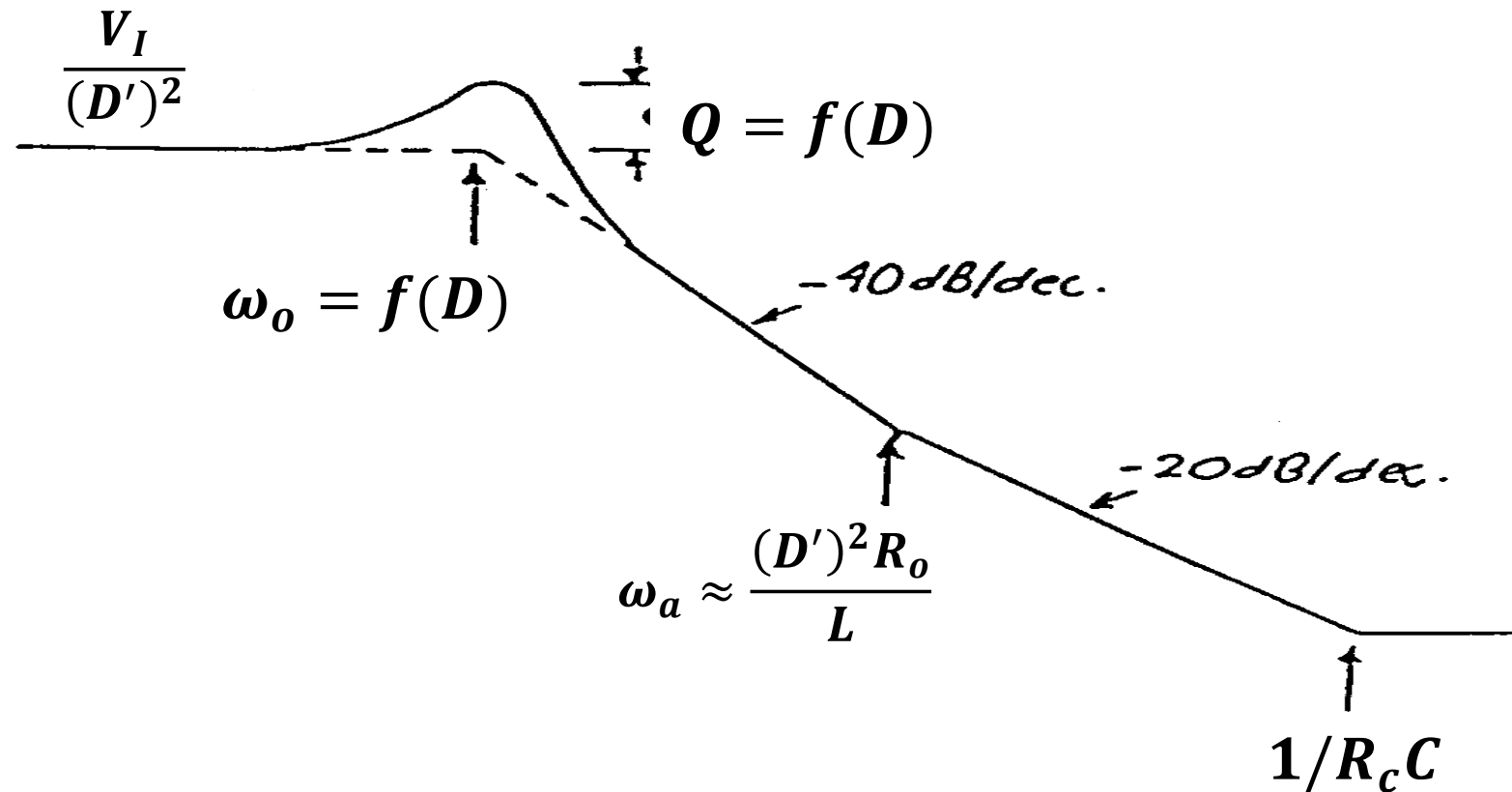
$$\frac{\hat{v}_o}{\hat{d}} = - \frac{V_I}{(D')^2} \frac{R_{le}}{R_o + R_{le}} \frac{(1 + sCR_c)(1 + sL_e/R_{le})}{\Delta(s)}$$

Total effect: $\frac{\hat{v}_o}{\hat{d}} \approx \frac{V_I}{(D')^2} \frac{(1 + sCR_c)(1 - s/\omega_a)}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}}$ **Right-half-plane (RHP) zero:** $\omega_a \approx \frac{(D')^2 R_o}{L}$

Open-Loop Duty-Cycle-to-Output Transfer Function

The asymptotic magnitude Bode plot is shown below:

$$\frac{\hat{v}_o}{\hat{d}} \approx \frac{V_I}{(D')^2} \frac{(1 + sCR_c)(1 - s/\omega_a)}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}} \quad \omega_0 \approx \frac{D'}{\sqrt{LC}} \quad Q \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{(D')^2 R_o} + C \left(R_c + \frac{R_l}{(D')^2} \right)}$$

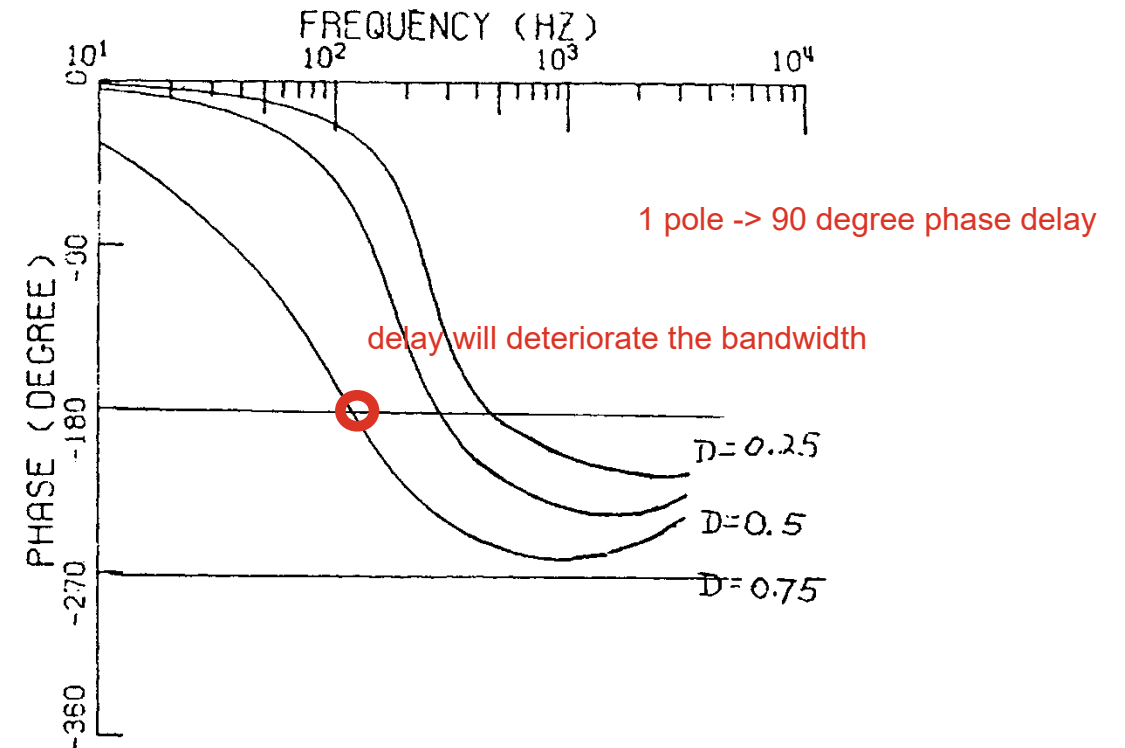
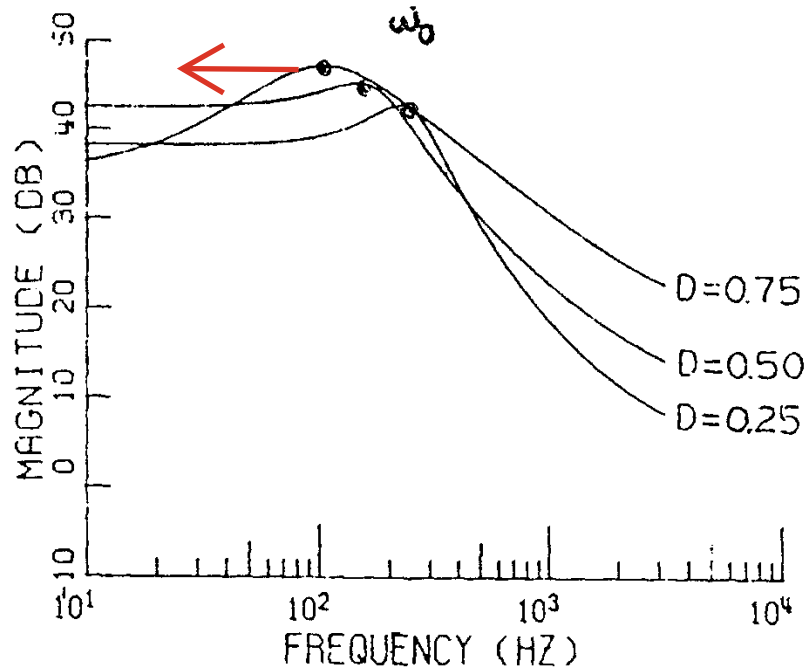


Open-Loop Duty-Cycle-to-Output Transfer Function

RHP will decrease phase but still increase gain
which means more delay in system

$$\frac{\hat{v}_o}{\hat{d}} \approx \frac{V_I}{(D')^2} \frac{(1 + sCR_c)(1 - s/\omega_a)}{1 + \frac{1}{Q} \frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}} \quad \omega_0 \approx \frac{D'}{\sqrt{LC}}$$

$$Q \approx \frac{1}{\omega_0} \frac{1}{\frac{L}{(D')^2 R_o} + C \left(R_c + \frac{R_{le}}{(D')^2} \right)} \quad \omega_a \approx \frac{(D')^2 R_o}{L}$$



Moving Poles
Moving Right-half-Plane Zero